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CXR-MultiTaskNet a unified deep learning framework for joint disease localization and classification in chest radiographs

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Chest X-ray (CXR) is a challenging problem in automated medical diagnosis, where complex visual patterns of thoracic diseases must be precisely identified through multi-label classification and lesion localization. Current approaches typically consider classification and localization in isolation, resulting in a piecemeal system that does not exploit common representations and is often not clinically interpretable, as well as limited in handling multi-label diseases. Although multi-task learning frameworks, such as DeepChest and CLN, appear to meet this goal, they suffer from task interference and poor explainability, which limits their practical application in real-world clinical workflows. To address these limitations, we present a unified multi-task deep learning framework, CXR-MultiTaskNet, for simultaneously classifying thoracic diseases and localizing lesions in chest X-rays. Our framework comprises a standard ResNet50 feature extractor, two task-specific heads for multi-task learning, and a Grad-CAM-based explainability module that provides accurate predictions and enhances clinical explainability. We formulate a joint loss that weighs the relative importance of representation extraction, which is large due to class variations, and the final loss, which is larger in the detection loss that occurs in extreme class imbalances between days and the detectability of varying disease manifestation types. Recent advances made by deep learning methods in the identification of disease in chest X-ray images are promising; however, there are limitations in their performance for complete analysis due to the lack of interpretability, some inherent weaknesses of convolutional neural networks (CNN), and prior learning of classification at the image level before localization of the disease. In this paper, we propose a dual-attention-based hierarchical feature extraction approach, which addresses the challenges of deep learning in detecting diseases in chest X-ray images. Through the use of visual attention maps, the detection steps can be better tracked, and therefore, the entire process is made more interpretable than with a traditional CNN-embedding model. We also manage to obtain both disease-level and pixel-level predictions, which enable explainable and comprehensive analysis of each image and aid in localizing each detected abnormality area. The proposed approach was further optimized for X-ray images by computing the objective losses during training, which ultimately gives higher significance to smaller lesions. Experimental evaluations on a benchmark chest X-ray dataset demonstrate the potential of the proposed approach achieving a macro F1-score of 0.965 (0.968 micro F1-score) for disease classification and mean IoU of 0.851 (mAP@0.50) for localization of diseases Content: Model interpretability, Chest X-ray image disease detection, Detection region localization, Weakly supervised transfer learning Lesion localization → 5 of 0.927 Compared to state-of-the-art single-task and multi-task baselines, these results are consistently better. The presented framework provides an integrated, method-based approach to chest X-ray analysis that is clinically useful, interpretable, and scalable for automation, allowing for efficient diagnostic pathways and enhanced clinical decision-making. This single framework can serve as a router for next-gen explainable AI in radiology.

Keywords Chest X-ray analysis, Multi-task learning, Disease classification, Lesion localization, Explainable deep learning

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Chest X-ray (CXR) interpretation is crucial for the early detection and monitoring of thoracic diseases. Yet, it remains challenging for automated analysis due to underlying anatomical structures, heterogeneous representation of the disease, and the multi-label nature of the prediction. Although traditional deep learning frameworks have made significant progress in either disease classification or lesion localization, few of them provide a unified approach that can accomplish both tasks simultaneously. Previously proposed models, such as DeepChest¹ and CLN², focused on multi-task learning (MTL) but did not leverage explainability or support multi-disease tasks. Likewise, while most of the above works employed shared learning techniques^{4,3}, their task diversity was limited, which reduced their generalizability in real-world clinical applications.

CXR-MultiTaskNet: A multi-task deep learning framework for chest radiograph interpretation, which addresses the limitations above by taking CNR250K and other datasets to solve the tasks of thoracic disease classification and localisation in a unified approach. We propose a shared feature extraction approach, utilizing a ResNet50 backbone, optimized dual-task heads, and Grad-CAM-based explainability. This design mitigates the operational inefficiency incurred by single-task pipelines and isolated feature learning identified in prior works, whilst outputting interpretable results essential for clinical uptake.

The core novelties of the proposed system include balanced multi-task optimization to avoid task interference, a joint loss formulation that accounts for both detection and classification objectives, and a clinically meaningful explainability module integrated into the diagnostic pipeline. These innovations enable CXR-MultiTaskNet to generalize well across varying disease presentations and class imbalances, addressing key gaps identified in prior studies.

In contrast to the current status quo in the form of amalgamated classification and detection heads, CXR-MultiTaskNet is explicitly developed concerning a host of integral challenges associated with chest X-ray analysis: (i) using an adequately balanced joint loss formulation to mitigate task dependency during co-learning; (ii) accommodating multi-label manifestation encompassing different diseases via optimal multi-task training; (iii) univocally incorporating a clinically coherent Grad-CAM module generating lesion-specific saliency maps registered onto detected lesions; and (iv) proficiently learning from datasets enriched jointly by weak (image-level) as well as strong annotations (bounding-box level). With these design considerations, the proposed system can deliver strong performance across prevalent and uncommon pathologies while retaining high clinical interpretability.

This paper makes the following contributions: (i) a unified architecture for simultaneous multi-label classification and lesion localization, (ii) a task-balancing optimization strategy to enhance model generalization and robustness, (iii) an integrated explainability mechanism for clinical interpretability, and (iv) extensive evaluation on public chest X-ray datasets demonstrating significant performance improvements over state-of-the-art methods.

The remainder of this paper is organized as follows: “[Related work](#)” reviews related work on multi-task learning, chest X-ray analysis, and explainable AI. “[Preliminaries](#)” presents the preliminaries covering background on chest X-ray disease diagnosis, multi-task learning concepts, dataset details, and problem formulation. “[Proposed system](#)” details the proposed CXR-MultiTaskNet architecture, including data preprocessing, model design, loss functions, and explainability modules. “[Experimental results](#)” presents the experimental setup, quantitative and qualitative results, and ablation studies. “[Discussion](#)” provides a comprehensive discussion and outlines limitations of the study in “[Limitations of the study](#)”. Finally, “[Conclusion and future work](#)” concludes the paper and highlights future research directions to improve further clinical adoption and scalability of automated chest X-ray analysis systems.

Related work

Recent advancements in deep learning have significantly improved chest X-ray analysis, particularly in disease localization and classification. This section reviews key contributions across multi-task learning frameworks, transformer-based models, and explainable AI techniques. It further highlights efforts in addressing domain adaptation, data imbalance, and noisy labels, providing a comprehensive background for the proposed CXR-MultiTaskNet framework.

Multi-task learning frameworks for chest X-ray analysis

Recent developments in multi-task learning frameworks have further advanced chest X-ray analysis by enabling the joint modeling of disease classification and localization, along with related auxiliary tasks, thereby improving model efficiency and clinical relevance. Mohamed et al. DeepChest¹ presented a model-agnostic multi-task learning (MTL) approach that adaptively assigns task weights, achieving both efficiency and accuracy better than all previous ones. Similarly, Okolo et al. CLN, a dual-branch architecture achieving impressive classification (AUC 0.918) and localization (IoU 0.855) on ChestX-ray14². Dealing with resolution limitations, Akhter et al. To facilitate diagnosis in limited-resource settings³, developed the MLCaK method, which learns to transfer knowledge from high-resolution to low-resolution CXRs. Zhu et al. They employed a Multi-task UNet architecture to train a saliency model and a disease classifier jointly, yielding benefits for both interpretability and diagnosis accuracy⁴.

Ai et al. tackled pneumonia diagnosis. Specifically⁵, proposed a multiple-task learning (MTL) framework that integrates patient-specific demographics, such as age, gender, and diabetes duration, as features. Al Zahrani et al. An MTL model focused on detecting thoracic pathologies leveraged the simultaneous understanding of pneumonia and COVID-19, as described in⁶. In addition to these works, Kamiri et al. (A systematic review of MTL in medical imaging⁷ identified several potential research gaps, including the combination of multiple

tasks into a single one and irregularities in results due to generalizability issues. Lastly, Qiu et al.¹⁰ proposed a cooperative MTL framework, where both radiological gaze prediction and diagnosis were compared in an explainable and clinically visionary manner.

Xu et al. Multi-task transformer model that has been tuned with unified instructions to achieve strong performance in classification, localization, and report generation on a large-scale collection of chest radiographs¹³. Li et al.¹⁴ presented a novel work that constructed the RMT (Robust Multimodal Transformer) framework to jointly leverage X-ray and clinical text data for assessing pneumonia presence in pediatric patients, which is more capable of handling missing modalities and appears to be robust. Ghamizi et al. Improving model generalization across datasets was reported in¹⁵, which demonstrated that learning auxiliary tasks in MTL frameworks improves cross-domain chest disease classification. Lin et al. With the approach of residual learning and MTL combined¹⁶, lung nodules detection and classification were enhanced using Res-MTNet.

Zhang et al. CXR-Net — A neural network for explainable and accurate COVID-19 pneumonia diagnosis, combining classification and explainability modules⁶⁵. Park et al. The authors of⁶⁶ present a novel vision and language approach called M4CXR, which incorporates multi-tasking capabilities into a large multimodal language model for chest X-ray interpretation, encompassing both report generation and image captioning for the M4CXR component. Zhu et al. In fine-grained tasks, leveraging a multi-task transformer⁶⁷ has been used to label chest X-ray reports, facilitating the automatic extraction of various diagnostic terms. Liao et al. An approach to improving X-ray image analysis through shared representation learning across tasks: MTPRet⁶⁸.

Wang et al.⁶⁹ applied a multi-task learning framework to predict central venous catheter status from radiographs, addressing both detection and classification within a single model. Finally, Ullah et al.⁷⁰ proposed a dual encoder–decoder deep learning framework for anatomical structure segmentation in chest X-rays, improving feature localization and segmentation accuracy. Tan et al.⁷³ presented DeepPulmoTB, a benchmark dataset that supports multi-task learning of tuberculosis lesions in lung CT images, facilitating both lesion segmentation and classification. Gende et al.⁷⁹ proposed an end-to-end multi-task framework for simultaneous epiretinal membrane segmentation and screening in OCT images, highlighting the broader application of multi-task learning in medical imaging.

Transformer-based, self-supervised, and multi-modal learning in medical imaging

Transformer architectures and self-supervised learning have transformed radiograph interpretation by capturing complex dependencies, leveraging unlabeled data, and integrating multimodal clinical information. Liao et al.¹¹ proposed MUSCLE, a self-supervised continual learning framework that pre-trains models across multiple X-ray datasets, addressing data heterogeneity and enhancing transferability for classification and segmentation. Dong et al.¹² designed a multi-channel MTL framework to predict EGFR and KRAS mutations in lung cancer using CT images, demonstrating the clinical potential of combining imaging and patient data. Xiao et al.⁴¹ explored masked autoencoders for multi-label classification of thoracic diseases, demonstrating improved feature learning and robustness in chest X-ray analysis. Li et al.⁴² introduced a visual-semantic embedded knowledge graph to enhance multi-label learning, improving the contextual understanding of radiology images. Huang et al.⁴³ proposed a two-stage training framework with label decoupling and reconstruction to address the challenges of long-tailed multi-label medical image recognition.

Taslimi et al.⁴⁴ developed SwinCheX, a transformer-based architecture for multi-label thoracic disease detection, emphasizing the benefits of hierarchical attention mechanisms. Chen et al.⁴⁵ introduced BOMD, a bag-of-multi-label descriptors framework, improving noisy chest X-ray classification through advanced representation learning. Hayat et al.⁴⁶ applied generalized zero-shot learning for chest radiograph classification, enabling the recognition of unseen diseases by leveraging semantic relationships. Wang et al.⁵⁵ proposed a multi-granularity cross-modal alignment approach for generalizable medical visual representation learning, enhancing the interpretability of chest X-ray analysis. Zhou et al.⁵⁶ advanced radiograph representation learning using masked record modeling, which integrates textual clinical records with visual features for improved disease prediction.

Ye et al.⁵⁷ introduced continual self-supervised learning for universal multi-modal medical data representation, facilitating cross-domain transfer in chest radiograph analysis. Gündel et al.⁵⁸ proposed integrating external knowledge sources to improve chest abnormality classification under noisy labels, enhancing robustness and generalization. Nguyen et al.⁵⁹ focused on multi-language radiology report mining, building a deep learning model for thoracic disease diagnosis from Vietnamese chest radiographs. Jang et al.⁶⁰ significantly improved zero-shot pathology classification on chest X-rays by fine-tuning image-text encoders, enabling scalable detection of unseen diseases. Seibold et al.⁶¹ introduced a report-guided contrastive training approach that allows chest X-ray models to recognize a flexible range of pathologies by aligning image features with the semantics of clinical reports, thereby moving beyond fixed-label classification.

Gazda et al.⁶² proposed a self-supervised CNN framework for chest X-ray classification, significantly reducing annotation requirements while achieving strong diagnostic performance. Zhao et al.⁶³ presented a segment-enhanced contrastive learning approach to improve medical report generation, facilitating better alignment between image regions and textual descriptions. Vu et al.⁶⁴ developed MedAug, a contrastive learning framework that leverages patient metadata to improve chest X-ray representations, thereby enhancing model robustness and transferability. Sirbu et al.⁷¹ introduced GIT-CXR, a transformer-based end-to-end framework that simultaneously generates chest X-ray diagnostic reports, enhancing clinical interpretability while also facilitating pathology detection. Yu and Zhou⁷² optimized a transformer model with multi-scale feature fusion, improving the detection efficiency of thoracic diseases in chest X-rays.

Multi-label and multi-class chest disease classification

Deep learning-based classification approaches have focused on handling the multi-label nature of chest X-ray datasets, addressing thoracic diseases with models leveraging residual learning, ensemble methods, and domain-specific optimizations. Su et al.⁸ explored multi-view MTL for CT image quality assessment, extending MTL beyond classification. Kumar et al.⁹ applied ensemble learning with preprocessing and feature optimization for the rapid diagnosis of pulmonary diseases. Alshmrani et al.¹⁷ presented a deep learning model for multi-class lung disease classification using CXR, achieving competitive accuracy across multiple conditions. Kim et al.¹⁸ utilized EfficientNet for end-to-end multi-class lung disease classification, emphasizing the importance of transfer learning in enhancing diagnostic efficiency. Emara et al.¹⁹ combined super-resolution and classification using InceptionResNetv2, improving the quality and accuracy of lung disease detection on low-resolution scans.

Lastly, Shaheed et al.²⁰ proposed an optimized Xception model with XGBoost for multiclass chest disease classification, improving computational efficiency and diagnostic accuracy in resource-constrained environments. In the pursuit of high-precision lung disease detection, Shamrat et al.²¹ introduced MobileLungNetV2, a fine-tuned MobileNetV2 model that achieved remarkable classification accuracy for 14 lung conditions using the ChestX-ray14 dataset. Deepak and Bhat²² proposed a multi-stage deep learning pipeline for comprehensive lung disease classification, integrating data augmentation and multi-scale learning for improved generalization. Li et al.²³ presented a Multi-Level Residual Feature Fusion Network, effectively enhancing thoracic disease classification through deep residual learning and multi-scale feature extraction. Chowdary and Kanhangad³¹ introduced a dual-branch network that integrates global and local features for thoracic disease diagnosis, enhancing multi-label classification performance on chest X-rays.

Malik et al.³² combined chest X-rays, CT scans, and cough sound images in a deep learning framework, demonstrating a novel multimodal approach for classifying chest diseases. Xu et al.³³ proposed MS-ANet, an automated multi-label detection system using attention-based networks for thoracic disease classification, achieving superior results on multiple datasets. Bhusal and Panday³⁴ utilized DenseNet for multi-label classification, emphasizing the need for robust feature extraction in detecting multiple thoracic abnormalities. Holste et al.³⁵ discussed the challenges of long-tailed disease distributions in chest X-rays and introduced the CXR-LT challenge to benchmark models on real-world imbalance scenarios. Lai et al.³⁶ addressed noisy labels and class imbalance in long-tailed, multi-label classification using advanced label correction and class distribution modeling techniques.

Wang et al.³⁷ introduced a cross- and intra-image prototypical learning method, enhancing interpretability and multi-label classification accuracy on thoracic images. Kufel et al.³⁸ applied transfer learning for chest X-ray abnormality classification, showcasing improved generalization through fine-tuned pre-trained models. Öztürk et al.³⁹ proposed HydraViT, a transformer-based adaptive multi-branch model for thoracic disease classification, outperforming CNN-based models in capturing complex feature dependencies. Efimovich et al.⁴⁰ integrated deep learning with natural language processing for multilabel classification of lung diseases, facilitating better alignment between radiological images and textual reports for enhanced clinical interpretability. Ukwuoma et al.⁷⁶ designed a hybrid explainable ensemble transformer encoder for pneumonia detection from chest X-rays, combining performance with interpretability.

Explainability, visual attention, and clinical interpretability

Explainable AI techniques, such as saliency mapping, attention mechanisms, and anatomically consistent embedding, have been integrated into chest X-ray models to improve interpretability and clinical trust. Elhanashi et al.²⁴ combined classification and localization tasks using a unified network, demonstrating improved interpretability for multi-type chest X-ray abnormalities. Mochurad et al.²⁵ applied CNN-based methods for basic chest abnormality classification, emphasizing efficient feature learning from X-ray images. Sajed et al.²⁶ provided a systematic review comparing deep learning and traditional machine learning for lung disease diagnosis, highlighting the superior performance of deep architectures in recent studies. Azad et al.²⁷ explored multi-disease detection combining ML and DL techniques, emphasizing lightweight frameworks for resource-limited settings. El Asnaoui et al.²⁸ reviewed automated pneumonia detection, underscoring the need for better annotated datasets and explainable models.

Liu et al.²⁹ proposed a deep learning model for differentiating TB and non-tuberculous mycobacterial lung disease (NTM-LD) using chest X-rays, validated through a clinical cross-sectional study. Wang et al.³⁰ demonstrated the multi-site applicability of AI models in diagnosing chronic obstructive pulmonary disease (COPD) from chest X-rays, emphasizing the role of cross-modal and multi-institutional learning frameworks. Cheng et al.⁴⁷ proposed a deep learning-based object detection strategy to simultaneously classify and localize thoracic abnormalities, thereby enhancing interpretability and diagnostic accuracy. Mostafa et al.⁴⁸ presented a comprehensive survey of AI techniques for thoracic disease diagnosis, highlighting trends in deep learning, transfer learning, and explainability. Li et al.⁴⁹ focused on interpretable thoracic pathology prediction through group-disentangled representation learning, improving transparency in disease classification.

Zhou et al.⁵⁰ developed an anatomically consistent embedding framework to enhance radiograph interpretation, aligning feature learning with human anatomical structures for better explainability. Celniak et al.⁵¹ improved thoracic radiograph classification through inter-species and inter-pathology self-supervised pre-training, demonstrating the transferability of learned features across veterinary and human datasets. Tran et al.⁵² developed a deep CNN model for diagnosing multiple diseases in pediatric chest radiographs, achieving high accuracy by learning from noisy and incomplete labels. Tiu et al.⁵³ demonstrated expert-level detection of chest pathologies using self-supervised learning on unannotated X-rays, highlighting the scalability of unsupervised feature learning. Van der Sluijs et al.⁵⁴ explored image augmentations in Siamese networks for chest X-ray representation learning, improving disease classification performance with contrastive learning techniques.

Lian et al.⁷⁴ proposed a DRD U-Net integrated with WGAN and deep neural networks for lung image segmentation, demonstrating improved boundary delineation in chest CT scans. Xiao and Zhao⁷⁵ developed a multi-task conditional GAN for synthesizing 3D liver CT images from vascular structures, illustrating the versatility of generative models in medical imaging. Sharmay et al.⁷⁷ explored transfer learning in histopathology through the HistoTransfer framework, providing insights applicable to chest X-ray domain adaptation. Feng et al.⁷⁸ applied deep supervised domain adaptation to improve pneumonia diagnosis across chest X-ray datasets with distributional differences. Lastly, Lee and Liu⁸⁰ demonstrated a real-time end-to-end deep learning framework for lane detection and path prediction in autonomous driving, illustrating the effectiveness of multi-task learning beyond healthcare.

Robustness against data challenges: domain adaptation, imbalanced datasets, and label noise

Studies have addressed critical challenges such as domain shifts, noisy annotations, and imbalanced class distributions, proposing domain adaptation, noise-robust training, and augmentation strategies to enhance generalization. Holste et al.³⁵ discussed the challenges of long-tailed disease distributions in chest X-rays and introduced the CXR-LT challenge to benchmark models on real-world imbalance scenarios. Lai et al.³⁶ addressed noisy labels and class imbalance in long-tailed, multi-label classification using advanced label correction and class distribution modeling techniques. Wang et al.³⁷ introduced a cross- and intra-image prototypical learning method, enhancing interpretability and multi-label classification accuracy on thoracic images. Kufel et al.³⁸ applied transfer learning for chest X-ray abnormality classification, showcasing improved generalization through fine-tuned pre-trained models. Öztürk et al.³⁹ proposed HydraViT, a transformer-based adaptive multi-branch model for thoracic disease classification, outperforming CNN-based models in capturing complex feature dependencies.

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Table 1 summarizes key multi-task chest X-ray studies, highlighting their contributions and identifying research gaps addressed by the proposed framework. The reviewed literature demonstrates significant progress in multi-task learning, transformer-based models, and explainable AI for chest X-ray analysis. However, existing approaches often lack a unified framework that efficiently integrates disease localization and classification while ensuring clinical interpretability. Addressing these gaps, the proposed CXR-MultiTaskNet leverages joint learning and explainable modules to deliver accurate and transparent chest radiograph analysis.

Preliminaries

Chest X-ray disease diagnosis and challenges

Chest radiography (CXR) is one of the most frequently performed imaging examinations. It is relied upon to diagnose various pulmonary and vascular diseases because of its low cost, short performance time, and ready availability. It is widely used in the detection and monitoring of diseases including pneumonia, pulmonary edema, cardiomegaly, atresia, tuberculosis, and pleural effusion. Nevertheless, although CXR has become a clinical commodity, the visual analysis of CXR images can be challenging due to the subtle appearances of specific pathologies, a large degree of anatomical overlap, and heterogeneous imaging quality between different acquisitions. Furthermore, multiple anomalies are present in a single radiograph, thus indicating that the diagnosis is multi-label. The variability in human interpretation is also put forth as a challenge, with a high level of inter-observer disagreement seen even among expert radiologists (2). These challenges are compounded by the absence of large numbers of highly accurate, well-labelled datasets, in which a disease is labeled as present and its exact position identified, since labeling both the presence of the disease and the precise location of the lesion is costly. Accordingly, robust models with learned weakly supervised information, imbalanced data

Author(s), year	Approach/model	Dataset	Key contributions	Research gap aligned to proposed system
Mohamed et al. ¹	DeepChest (MTL)	ChestX-ray14	Dynamic task weighting in MTL improves efficiency and accuracy	Does not integrate explainability or joint lesion localization
Okolo et al. ²	CLN (Dual Branch MTL)	ChestX-ray14	Achieved high AUC and IoU in classification and localization	Limited to dual-task setup, lacks multi-label disease complexity
Akhter et al. ³	MLCAK (MTL Knowledge Transfer)	Private CXR Dataset	Knowledge transfer across resolutions for better diagnosis	Focus on resolution transfer, lacks multi-label detection generalizability
Zhu et al. ⁴	Multi-task UNet	NIH CXR	Combines saliency prediction and classification to improve interpretability	Does not perform bounding box localization, limited task diversity
Zhang et al. ⁶⁵	CXR-Net (Explainable MTL)	COVID-19 CXRs	Combining explainability modules with classification for pneumonia	Focused on COVID-19 only, not generalized to multiple thoracic diseases
Xu et al. ¹³	Multi-Task Transformer	Multiple CXR Tasks	Unified instruction-tuned transformer for classification, localization, and report generation	Computationally heavy; lacks task balancing and explainability integration.
Liao et al. ⁶⁸	MTPRet (Multi-Task Pretraining)	Public X-ray datasets	Multi-task pretraining for shared representation learning	Does not directly address real-time multi-label localization and classification
Wang et al. ⁶⁹	Multi-Task Central Venous Catheter Prediction	Radiograph datasets	Single-model MTL for catheter detection and classification	Task-specific; lacks general thoracic disease applicability and explainable outputs

Table 1. Summary of highly relevant literature and identified research Gaps.

distributions, and generalization across heterogeneous clinical scenarios are essential for automated systems to diagnose disease and localise its focus simultaneously.

Multi-task learning in medical imaging

Multi-task learning (MTL) is a method that trains a model to perform more than one task, using shared representations to improve the overall effectiveness and generalization. For medical imaging, MTL has recently become popular because it allows the capture of complementary information between different diagnosis tasks. Contrary to single-task networks that only learn task-specific mappings, MTL networks enable the learning of shared feature representation across multiple functions that may be adapted to disease classification, lesion detection, segmentation, and severity scoring.

For CXR analysis, MTL methods are instrumental since many radiological findings have global as well as localizing presentations. For instance, for the diagnosis of cardiomegaly, a global shape deformation is being detected, whereas nodules and effusions call for an accurate spatial reasoning. In that way, by jointly performing classification and detection in a single framework, the model can take full advantage of the mutual influence between global diagnostic information and local visual hints. This standard feature learning process can alleviate overfitting and guarantee the efficiency of sample utilization, particularly when labeled samples are scarce.

Let I represent an image of a chest X-ray. Specifically, we wish to learn a function $f_{\theta}(I)$ parametrized by θ , which takes a disease label vector $\hat{y} \in \{0,1\}^C$ and a set of bounding box predictions $\hat{B} = \{(x_i, y_i, w_i, h_i, c_i)\}_{i=1}^N$ as inputs. The loss of MTL usually combines with classification and detection loss as follows:

$$\mathcal{L}_{total} = \lambda_1 \mathcal{L}_{cls} + \lambda_2 \mathcal{L}_{det} + \lambda_3 \mathcal{L}_{conf} \tag{1}$$

Here, $\mathcal{L}_{cls}$ is the binary cross-entropy loss for multi-label disease classification, $\mathcal{L}_{det}$ is the smooth L1 loss for bounding box regression, $\mathcal{L}_{conf}$ and is the loss of confidence score prediction in detection. The hyperparameters $\lambda_1, \lambda_2, \lambda_3$ are for reweighing the importance of each task.

By jointly optimizing classification and localization objectives with shared feature backbones, MTL-based deep learning architectures have shown state-of-the-art performance across medical imaging benchmarks. Such systems are particularly adapted to chest X-rays, in which the appearance of the condition is spatially non-uniform and with respect to diagnosis. MTL allows the incorporation of auxiliary tasks including image reconstruction or anatomy structure localization, which additionally constrain the model towards medically-meaningful areas and enhance its robustness for real-world clinical practice.

Dataset description and structure

The dataset employed in this study is based on a curated, VinDr-CXR-inspired architecture of chest X-ray images annotated with respect to disease presence and localisation. For each case in the dataset there is a high-resolution PA CXR image and the corresponding set of annotations: a multi-label binary vector indicating the presence/absence of pathologies (collapsed over the labels invariant of the disease type), and a variable set of bounding boxes indicating the location of visible lesions or pathological abnormalities. With two-label structure, it allows multi-task learning, the classification task is to classify the global disease level and the detection task is to find and localize the corresponding pathological regions.

Let the dataset be defined as $D = \{(I_k, y_k, B_k)\}_{k=1}^M$, where I_k is the k -th chest X-ray image, $y_k \in \{0,1\}^C$ is the multi-label disease vector of possible disease classes for C and $B_k = \{(x_i, y_i, w_i, h_i, c_i)\}_{i=1}^{N_k}$ is the set of N_k bounding box annotations pertaining to the image. In each bounding box annotation, the top-left position of the bounding box (x_i, y_i) , the width w_i , the height h_i and the confidence score c_i of the bounding box in indicating the strength of the visual evidence, are provided.

The dataset consists of thousands of samples in variety of disease categories 3, including atelectasis, effusion, consolidation, cardiomegaly and infiltration. Disease frequency is highly imbalanced, with some diseases such as effusion is more common and the other are less frequent, such as mass or fibrosis. The diseases per image range from one to three on average, and the bounding box count per image varies across pathologies substantially. This asymmetry and inconsistency of label structure calls for learning strategies that can handle sparse supervision and different spatial granularities.

All images are resized to \$224 \times 224\$ pixels for training effectiveness and normalized using ImageNet statistics to take advantage of pretrained backbones. The covariance of the bounding box coordinates are correspondingly scaled to the size of the resized image. The original dataset is randomly split into training, validation, and testing subsets at an 80:10:10 ratio with similar class distribution maintained in each split. This diversified and balanced dataset allows for building a holistic deep learning model that simultaneously learns the disease-level classification along with spatial localization while respecting everyday clinical diversity.

Deep learning foundations

“Good” - Convolutional Neural Networks (CNNs) are the foundation of the emerging (pre)medical image analysis systems, due to their hierarchical way of extracting features. For interpreting chest X-ray, CNN is a good representation learner and is capable of capturing spatial patterns and texture details related to thoracic findings. For a given input image $I \in \mathbb{R}^{H \times W \times 3}$, a CNN-based encoder translates it into a high dimensional feature map, where h , w , and d are the height, width and depth of the encoded feature representation, respectively. Subsequently, features are passed into tasks specific head for classification and detection.

Transfer learning based on pretrained CNN models including ResNet has been a well-recognized routine in medical imaging due to the absence of well-annotated data. These are often initialized with weights pre-trained on other natural image datasets (e.g. ImageNet) and fine-tuned on the training set of the target medical domain. In this work, a ResNet as the shared backbone network directly followed by two parallel heads including one for multi-label disease classification and the other one for bounding box regression. The classification head uses global average pooling and a fully connected layer with sigmoid activation to estimate the probability of each disease class. The detection head processes the same feature map for regressing the bounding box coordinates and generate the corresponding confidence scores.

Formally, classification prediction is performed as: in (3) where represents dimension in the predicted classification vector $0 \ 1 \ 2 \ 3 \ 26,790 | 1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7 \ 8 \ 9 \ 10 \dots$ Fig.

$$\hat{y} = \sigma(W_c \cdot GAP(F) + b_c) \quad (2)$$

Where $GAP(F)$ is the global average pooling operation on the feature map F , W_c and b_c are the learnable weights and bias of the classification layer, and σ is the sigmoid activation function applied element-wise for multi-label prediction.

The feature map is fed into a convolutional sequence for the detection branch to predict the bounding box set:

$$\hat{B} = f_{det}(F) \quad (3)$$

Where f_{det} is the detection head predicting the box parameters (x, y, w, h) and a confidence score cc for each object proposal. Each task-specific head is optimized with its corresponding loss, following the MTL formulation as presented in Sect. 3.

CNNs, attention mechanisms, and hybrid networks (with Transformer blocks) to model long-range dependencies in medical images are new trends. However, in our work, we concentrate on CNN-based models, since they are known to be computationally efficient and effective in CXR interpretation. The common backbone with dual-task prediction readout allows end-to-end optimization while preserving task-specific flexibility.

Problem formulation

The objective of this study is to develop a unified multi-task deep learning model that can simultaneously classify diseases and localize lesions on chest X-rays. This is a dual-task architecture that enables the model to integrate global diagnostic knowledge with local spatial reasoning, which is crucial for comprehending complex thoracic pathologies. In contrast to traditional single-task pipelines, this method utilizes shared feature learning to enhance performance across multiple tasks, while also rendering predictions more interpretable.

For each case, we describe it as a triplet of (I_k, y_k, B_k) , where I_k is the input image y_k is a binary vector indicating existence of each of the disease class and B_k is a variable number of bounding-boxes to indicate lesion locations. These annotations supply complementary supervision signals, which helps the network learn representations that are semantically meaningful and spatially accurate.

The network f_θ is trained θ to optimize a composite loss function – a combination of classification, background regression and confidence prediction losses (Eq. (1)) using. The classification branch outputs a multi-label prediction vector $\hat{y}_k$, the the detection branch produces a set of predicted bounding boxes, each with an associated confidence score.

This formulation of the problem allows us to jointly optimize across heterogeneous tasks, which were previously modeled into simple models in this pipeline. Given the overlap of many pathologies or subtle patterns, the model can better leverage spatial relationships and exclude ambiguity by matching image-level disease presence with regional-level localization. The discriminative nature and transferability of learned features lay the foundation for using shared convolutional backbones with task-specific heads to substantially improve the accuracy and interpretability of diagnostic models in medicine.

Proposed system

Here, we introduce CXR-MultiTaskNet, a unified multi-task learning framework to jointly learn thoracic disease classification and lesion localization from chest X-rays. It captures input through shared feature extraction, task-specific learning heads, and interpretable results through Grad-CAM-based explainability. We elaborate on some of the key components of the proposed system, covering data preprocessing, model architecture, and optimization strategies.

System overview

We propose CXR-MultiTaskNet, a unifying deep learning framework for joint disease classification and lesion localization in chest X-ray automatic images. The overall system is depicted as a modular architecture in Fig. 1, consisting of data preprocessing, a shared feature extractor, two distinct heads for learning on the two tasks, and finally, modules for explainability and evaluation.

Far left: Preprocessing Pipeline — The workflow begins with chest X-ray images and corresponding annotations, which are processed through a preprocessing pipeline that standardizes image dimensions and scales bounding box coordinates. The processed images are passed through a convolutional neural network backbone, such as ResNet-50 in the current example, to extract deep feature representations that reflect both high-level and detailed patterns of disease.

The standard feature map is put into two parallel branches. This branch outputs multi-label disease classification probability scores for each disease class using a fully connected layer with a sigmoid activation function. In YOLO-style architectures, the network outputs bounding box coordinates and confidence scores using a detection head. This parallel architecture enables the model to extract complementary features, which help balance global semantic understanding with detailed spatial localization.

After the inference, a non-maximum suppression module refines the raw outputs by eliminating redundant bounding boxes. In a simultaneous process, the Grad-CAM-based explainability module generates visual heatmaps that highlight the parts of the X-ray images that indicate the presence of disease, facilitating clinicians' interpretation more easily.

Finally, the system's outputs include predicted disease labels, an adjusted bounding box for the best-labelled regions, and heatmaps for explainability, enabling accurate and interpretable diagnosis. Such an integrated approach enhances the clinical utility of automated analysis of chest X-ray imaging by allowing radiologists to make quick and confident diagnoses.

An overall flow of data and steps of the operation in the CXR-MultiTaskNet framework are illustrated in Fig. 2. In this workflow, we start with a collection of chest X-ray images and their corresponding annotations,

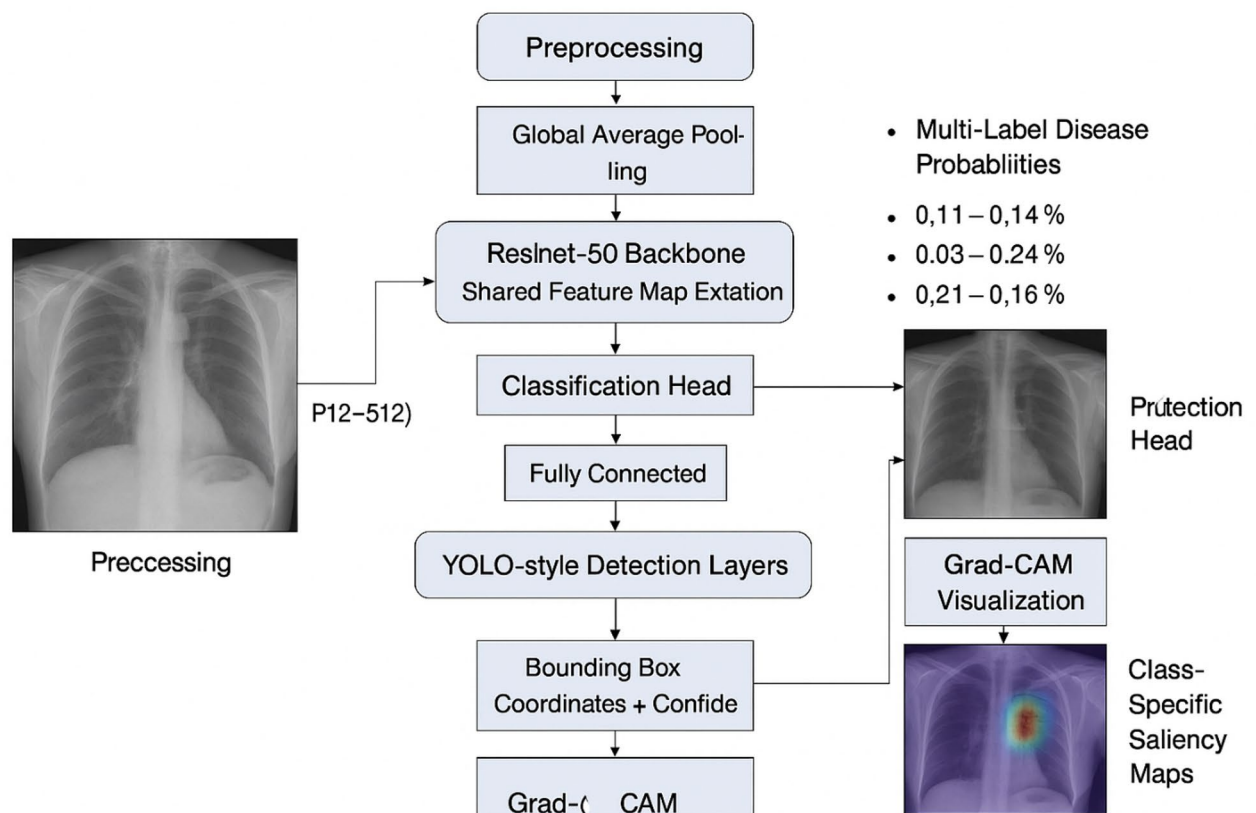


Fig. 1. System architecture of CXR-MultiTaskNet framework for simultaneous disease classification and lesion localization in chest X-rays.

which we first preprocess in a structured way. These preprocessing steps involve resizing all images to a consistent size, normalizing pixel intensities to conform to the prerequisites of the pre-trained models, and parsing the corresponding annotation files to retrieve disease labels along with their corresponding bounding box coordinates in normalized format.

After preprocessing, the input images are forwarded to the feature extraction module, where a ResNet-50 backbone is utilized to extract standard visual features that convey both global and local disease patterns. The same feature maps are then passed through two parallel branches. The first branch performs the multi-label disease classification, known as the classification head, which utilizes a fully connected layer and sigmoid activation to output disease probability scores. The detection head, corresponding to the second branch, is used to estimate bounding box coordinates and the confidence scores for spatial lesion localization.

Following the inference, the workflow proceeds to a bounding box post-processing stage, where Non-Maximum Suppression (NMS) is applied to remove redundant or overlapping detections, retaining only the most confident and distinct bounding boxes. In parallel, a Grad-CAM-based explainability module generates heatmaps by computing class-specific feature importance, thereby highlighting the disease-relevant regions in the original chest X-ray image.

The final stage of the workflow aggregates these outputs into clinically interpretable results, comprising predicted disease labels, refined bounding boxes, and Grad-CAM visual overlays. This end-to-end workflow supports efficient, explainable, and accurate automated analysis of chest X-rays, enhancing the diagnostic workflow in real-world clinical environments Table 2.

Data preprocessing and annotation handling

The proposed framework is based on the VinDr-CXR dataset, which consists of a large-scale, high-resolution poster-anterior Chest X-ray images with expert radiologists' annotations. Each image is associated with local annotations of thoracic abnormalities (if present, with a bounding box) and global labels of diagnostic results for six diseases (Pneumonia, Pneumothorax, COPD, Tuberculosis, Nodule, and Edema). The multitask learning that detects lesion locations and classifies disease categories simultaneously relies on these annotations.

To maintain consistency across the dataset and to reduce computation during model training, all images are converted from DICOM to RGB and resized to a fixed size of 512×512 pixels. We transform the range $[-1, 1]$ for each image tensor using the following mapping:

$$x' = \frac{x}{255} \times 2 - 1 \quad (4)$$

where x is the input pixel value, which is in the range $[0, 255]$ and x' is the output pixel value normalized. Normally, we can have a stable convergence without saturated gradients to the deeper layers of the network.

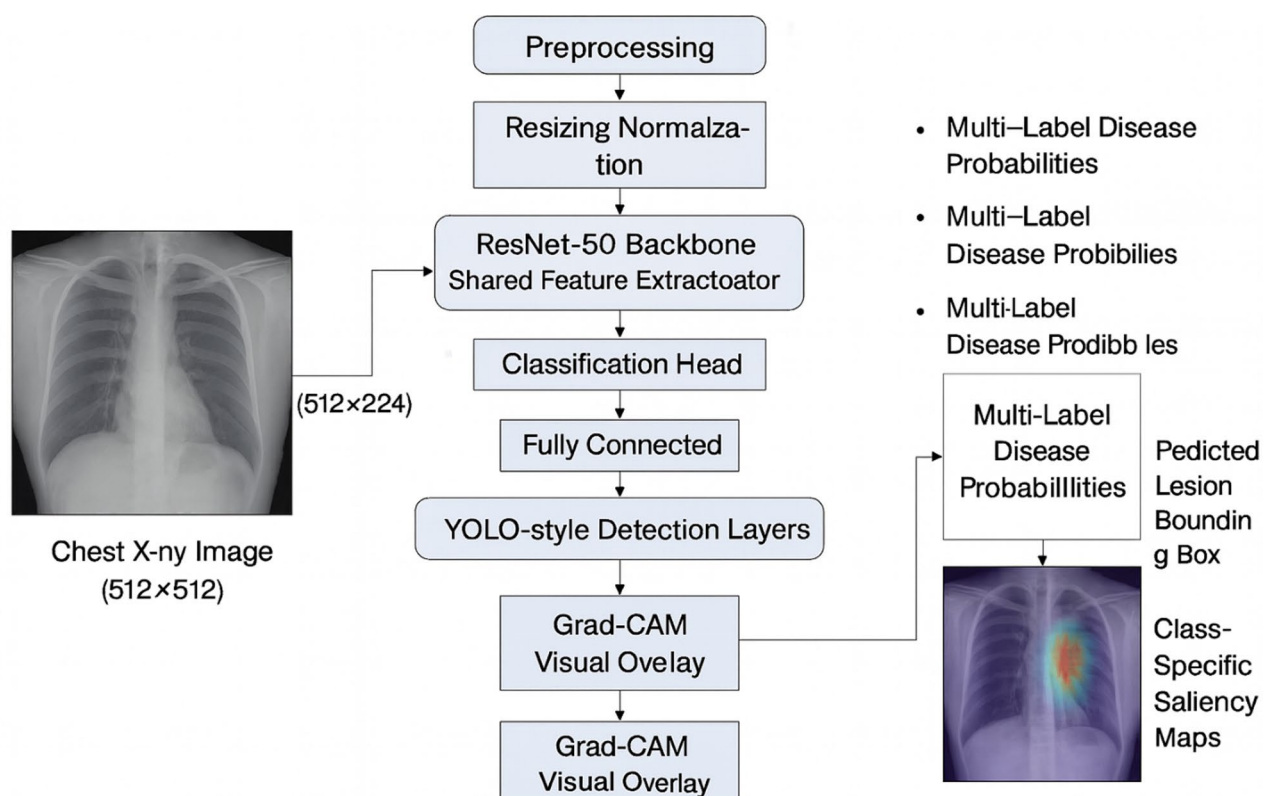


Fig. 2. End-to-end workflow of the proposed CXR-MultiTaskNet framework for chest X-ray disease analysis.

Symbol	Description
x_i	Input chest X-ray image for the i^{th} sample
y_i	Multi-label disease vector for x_i indicating presence of diseases
b_i	Set of ground-truth bounding boxes for x_i
$\hat{y}_i$	Predicted multi-label disease vector for x_i
$\hat{b}_i$	Predicted bounding boxes for x_i
θ	Trainable parameters of the multi-task model
θ^*	Optimized model parameters after training
f_θ	Multi-task model mapping input x_i to predictions
F	Shared feature map extracted from input image
α_k	Grad-CAM weight for the k^{th} channel
L_{cls}	Multi-label classification loss
L_{det}	Detection loss for bounding box regression and confidence
L_{total}	Combined loss: $L_{total} = \alpha L_{det} + \beta L_{cls}$
α, β	Task-specific weighting coefficients for loss balancing
$\hat{s}_c$	Predicted score (logit) for class c
H	Grad-CAM heatmap matrix for interpretability
τ	Confidence threshold for bounding box filtering
δ	IoU threshold for Non-Maximum Suppression (NMS)
N	Total number of training samples
B	Mini-batch size used in training
E	Number of training epochs
η	Learning rate for model optimization
IoU	Intersection-over-Union used for bounding box comparison

Table 2. Notations used in the proposed CXR-MultiTaskNet framework for multi-task chest X-ray disease analysis.

Local annotations are parsed from the local dataset to produce maximum three bounding boxes per images. Every bounding box is represented by (x, y, w, h, c) — where (x, y) are the coordinates of the center of the bounding box, (w, h) are its width and height, and c is a confidence score which is always 1.0 for ground-truths only. The detection targets are thus arranged into a fixed-shape tensor of size [B, N,5] where B is the batch size and N=3 is the maximum number of bounding boxes per image.

Simultaneously, the global diagnostic labels are represented as a binary vector of size 6 for each image, indicating the presence (1) or absence (0) of each disease category. Thus, multi-label classification is possible, and the model can predict multiple co-existing conditions for a single input instance.

The data for training and validation is split into three subsets — 70% for training, 15% for validation, and 15% for testing. Label distribution is preserved across splits to ensure good performance evaluation and prevent bias from class imbalance. The multi-task learning model finally accepts the processed dataset, leveraging these annotated structures to optimize both detection and classification objectives jointly.

Model architecture

The proposed CXR-MultiTaskNet model architecture is designed to share low- and mid-level feature representations for the common tasks of detection and classification of chest X-ray images. The model is based on the multi-task learning paradigm: it uses one backbone network to extract shared features, which are then analyzed by two different heads: one for bounding box regression and the other for multi-label classification.

Figure 3 illustrates the CXR-MultiTaskNet model architecture, where a ResNet-50 backbone extracts shared features, followed by parallel heads for multi-label disease classification and bounding box-based lesion localization. An input to the model is an RGB-colored chest X-ray image of size 512×512 with a 3-channel size. The inputs are fed into a pre-trained ResNet-50 architecture up to the last fully connected layer, retaining convolutional blocks that represent both low-level and high-level visual semantics. Specifically, the output of this backbone network provides a shared representation for both downstream tasks, a deep feature map of 2048×16×16.

The detection head operates in a YOLO-style manner to identify thoracic lesions. It is formed by a 3×3 convolutional layer and nature ReLU, at last one 1×1 convolutional layer to output a tensor size of the form [B, N, 5]: the size embankment size is B, N is the number of anchor boxes (we fixed N=3) and each anchor box predicts five values: center coordinates (x, y), width w height h and confidence score c together mathematically represented as:

$$B_i = (x_i, y_i, w_i, h_i, c_i), \quad i = 1, 2, \dots, N$$
 (5)

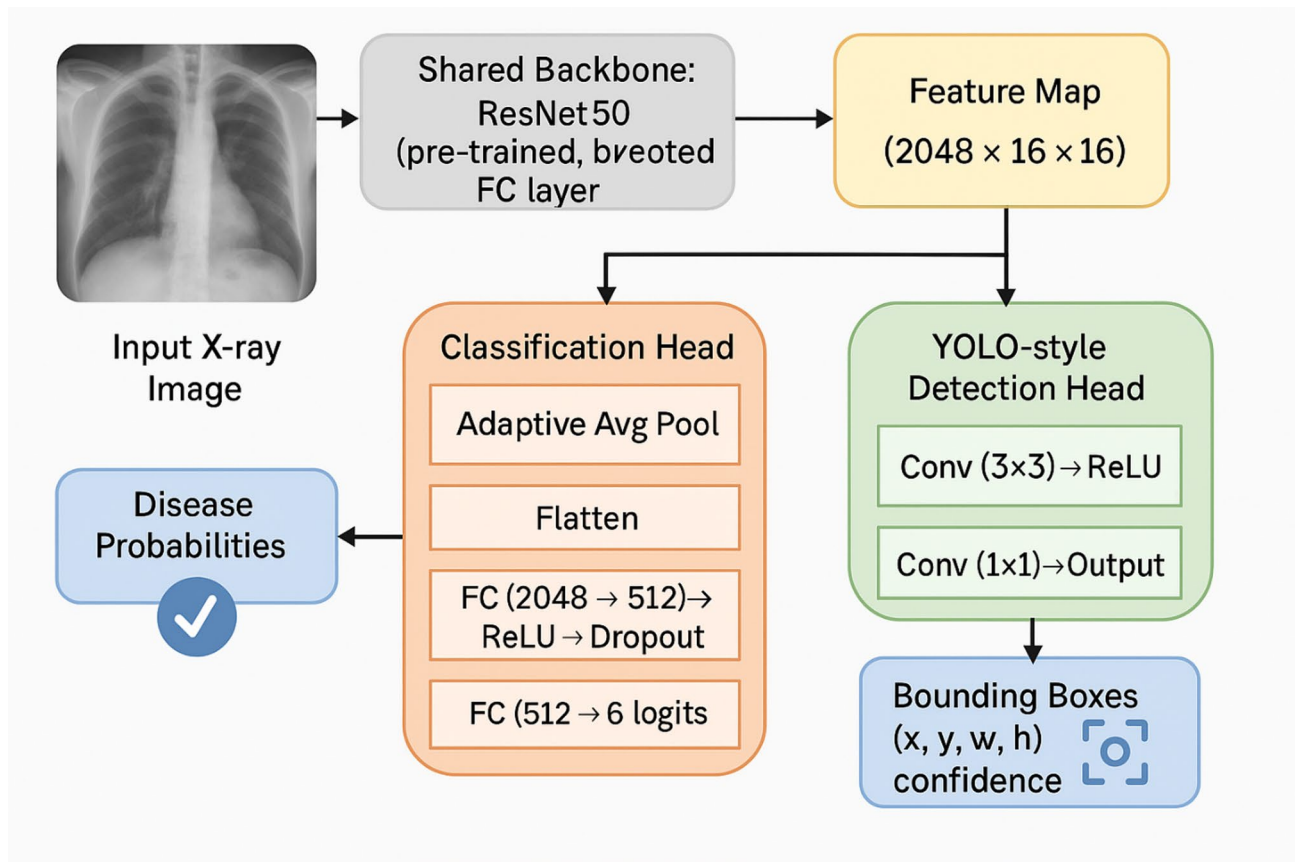


Fig. 3. Model architecture of CXR-MultiTaskNet showing shared feature extraction and dual-task prediction heads.

At the same time, the classification head is given the same shared feature map, which is then passed through a global average pooling layer, a flattening layer and two fully connected layers. The pooled feature vector is first passed through a fully connected layer that preserves the features across 2048 and 512 dimensions and applies a ReLU activation and dropout after each layer, followed by a final output layer of 6 logits that corresponds to the disease classes. The sigmoid activation is then applied to these logits to produce independent probability scores for each of the classes:

$$\hat{y}_j = \sigma(z_j), \quad j = 1, 2, \dots, 6 \quad (6)$$

logit(), where z_j is the logit for the j^{th} class and $\hat{y}_j$ is the predicted probability. By using this formulation, the model assumes that multi-label classification is necessary since multiple disease categories may be co-occurring in the same X-ray image.

CXR-MultiTaskNet allows end-to-end learning of both the spatial and categorical characteristics of disease by combining the outputs of a detection and classification head. By exploiting the similarities between localization and classification objectives, the shared backbone improves learning efficiency, alleviates parameter redundancy, and enables effective generalization across tasks.

Loss function formulation

The model CXR-MultiTaskNet is trained by a composite multi-task loss function that jointly minimizes object detection and multi-label classification. By taking this joint optimization approach, the model is pushed to learn common embeddings that can be useful for both tasks but still accounting for their specific learning targets via the task-specific loss components.

The total loss function L_{total} can be mathematically represented as a weighted sum of the detection loss L_{det} and the classification loss L_{cls} , as follows:

$$L_{total} = \alpha \cdot L_{det} + \beta \cdot L_{cls} \quad (7)$$

α, β are task-dependent weights (1.0 in this work).

The detection loss L_{det} is itself made up of two parts, one for regression (for coordinates of the bounding box) and the second for confidence (whether an object is present in the predicted region). This bounding box regression loss is defined as the Smooth L1 loss applied to the bounding box parameters (x, y, w, h) predicted

from the base network versus the ground-truth values, while the objectness confidence loss is a simple binary cross-entropy. The detection loss is formally expressed as:

$$L_{det} = \frac{1}{N} \sum_{i=1}^N \left[SmoothL1(\hat{b}_i, b_i) + BCE(\hat{c}_i, c_i) \right] \quad (8)$$

where $\hat{b}_i = (x_i, y_i, w_i, h_i)$ is the predicted bounding box, b_i is the ground-truth bounding box, $\hat{c}_i$ is the predicted confidence, and $c_i = 1.0$ for ground-truth boxes. N denotes the number of predicted boxes per image (up to 3).

The classification loss L_{cls} is defined as the binary cross-entropy loss of the multi-label outputs of the six disease categories. It punishes wrong predictions per class independently and supports multi-label predictions:

$$L_{cls} = \frac{1}{C} \sum_{j=1}^C BCE(\hat{y}_j, y_j) \quad (9)$$

Where $\hat{y}_j$ is the predicted probability for the class j , $y_j \in \{0,1\}$ is the ground-truth label, and $C=6$ the number of classes of diseases.

This multi-task loss formulation enables an end-to-end model capable of both learning to localize regions of interest and producing reliable inferences for the diagnosis. Together, this combination of objectives allows us to better match the real-world clinical scenario, where not only the location of the lesion is important, but also the semantic understanding of the disease category that would be critical for informing downstream clinical decisions.

Training and evaluation protocol

In the end-to-end learning process, the proposed CXR-MultiTaskNet model uses supervised means in which both detection and classification tasks are optimized together during training. It lets train the network on the VinDr-CXR dataset where are the image inputs $X \in \mathbb{R}^{3 \times 512 \times 512}$, and r is the given bounding-box annotations, and y the multi-label disease vector. We formulate the total multi-task loss L_{total} as follows to minimize:

$$L_{total} = \alpha \cdot L_{det} + \beta \cdot L_{cls} \quad (4 \text{ revisited}).$$

Specifically, where α and β are weights that balance the two tasks, and L_{det} , L_{cls} , are defined in Eqs. 5 and 6, respectively.

The model is trained using Adam optimizer with a learning rate of $\eta = 1 \times 10^{-4}$, and batch size of $B=8$. Only one learning rate is used for all epochs, no warm up, no decay schedule, 20 epochs are trained. Where θ is model parameters; training objective is the find optimal parameters θ^* that minimize the total loss over the training dataset:

$$\theta^* = \underset{\theta}{\operatorname{argmin}} \sum_{(x, b, y) \in D_{train}} L_{total}(f_{\theta}(x), b, y). \quad (10)$$

Where x is the input image, b is the ground-truth bounding box tensor, and y is the disease label vector.

When it comes to classification performance evaluation, the predicted logits z_j for each class j are sent through the sigmoid activation:

$$\hat{y}_j = \sigma(z_j) = \frac{1}{1 + e^{-z_j}}, \quad j = 1, 2, \dots, 6 \quad (11)$$

True binary predictions $\in \{0,1\}$ are obtained by applying a threshold $\tau=0.5$ on probabilities. Using these predictions the following evaluation metrics are calculated:

F1-score:

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (12)$$

Using continuous outputs $\hat{y}_j$, AUC (Area Under the ROC Curve) is calculated and pooled across the six classes.

Input images with the predicted boxes $B_i = (x_i, y_i, w_i, h_i, c_i)$ for $i=1, \dots, N$ that has been predicted $N=3$, for the detection head). Since we do not have pixel-level segmentation masks, visual overlay of predicted boxes gives us a qualitative understanding about spatial detection performance.

Grad-CAM (Gradient-weighted Class Activation Mapping) is a popular approach that achieves interpretability by identifying which part of the image is contributing more towards each class prediction using a combination of activation maps and their gradients with respect to the score of a particular class. Grad-CAM heatmap M_j for class j is calculated as; Eq. 13

$$M_j = \text{ReLU} \left(\sum_k \alpha_k^j A^k \right), \quad \alpha_k^j = \frac{1}{Z} \sum_i \sum_j \frac{\partial \hat{y}_j}{\partial A_{i,j}^k} \quad (13)$$

Where A^k is the k^{th} is the feature map from the last convolutional layer and Z is the number of pixels in each feature map. The typical visual interpretation of the heatmap is by upsampling and using it as an overlay on the input image. Together, this protocol enables strong training, predicting, and interpreting of models that are all necessary for the eventual incorporation of AI models into clinical practice.

Algorithmic implementation

This section presents the algorithmic realization of the proposed CXR-MultiTaskNet framework, detailing each computational step from data preprocessing to model training, inference, and post-processing. The implementation leverages multi-task optimization techniques to jointly learn disease classification and lesion localization tasks, while integrating explainability through Grad-CAM and refinement through Non-Maximum Suppression, ensuring interpretable and clinically relevant outputs.

Algorithm: Multi-Task Training Procedure for CXR-MultiTaskNet

Input:

Training dataset $D = \{(x_i, b_i, y_i)\}_{i=1}^N$

Model parameters θ , learning rate η , number of epochs E , batch size B

Output:

Optimized model parameters θ^*

1. Initialize model parameters θ
2. for epoch = 1 to E do
 3. Shuffle D and split into mini-batches of size B
 4. for each mini-batch $B = \{(x_j, b_j, y_j)\}_{j=1}^B$ do
 5. Compute shared feature map: $F \leftarrow \text{ResNet50}(x)$
 6. Predict bounding boxes: $b \leftarrow \text{DetectionHead}(F)$
 7. Predict class logits: $\hat{z} \leftarrow \text{ClassificationHead}(F)$
 8. Apply sigmoid: $\hat{y} = \sigma(\hat{z})$
 9. Compute detection loss L_{det}
 10. Compute classification loss L_{cls}
 11. Compute total loss: $L_{total} = \alpha \cdot L_{det} + \beta \cdot L_{cls}$
 12. Backpropagate gradients and update $\theta \leftarrow \theta - \eta \cdot \nabla_{\theta} L_{total}$
 13. end for
3. end for
4. Return θ^*

Algorithm 1. Multi-task training procedure for CXR-MultiTaskNet.

The end-to-end training procedure of the multi-task deep learning architecture is summarized in Algorithm 1, which is termed as CXR-MultiTaskNet, for disease classification and lesion localization in chest radiograph images. The training process starts by initializing the model parameters and iterating over the dataset for a fixed number of epochs. In each epoch, the dataset is shuffled and partitioned into mini-batches for stochastic and generalization properties. For each mini-batch, we compute shared feature representations from a deep convolutional backbone (such as ResNet-50). These features are then processed by two task-specific heads: the detection branch predicts bounding boxes, and the classification branch generates disease class logits.

Sigmoid activation is applied to the classification outputs to obtain multi-label probabilities. Then the algorithm calculates detection loss (e.g., localization, confidence), and classification loss (e.g., binary cross-entropy). That loss is combined with a comprehensive loss using TAK balancing weights. At the end, backpropagation was performed, and the model parameters were updated using a standard optimizer, such as SGD or Adam. The same procedure is repeated for all batches and epochs until the model converges to an optimal set of parameters, θ , which is provided as output in this case. Such a process allows for the joint optimization over the tasks, so that the model is capable to learn semantic and spatial disease patterns in an integrated way.

Algorithm: Bounding Box Post-Processing with Non-Maximum Suppression (NMS)**Input:** Predicted bounding boxes $\hat{B} = \{(x_i, y_i, w_i, h_i, c_i)\}_{i=1}^N$, confidence threshold τ , IoU threshold δ **Output:** Final refined bounding boxes B^*

1. Filter boxes with confidence $c_i < \tau$
2. Group remaining boxes by predicted class label
3. For each class:
 - a. Sort boxes by confidence in descending order
 - b. While box list is not empty:
 - i. Select box b_{max} with highest confidence
 - ii. Add b_{max} to B^*
 - iii. Remove all boxes with $IoU(b_{max}, b_j) > \delta$
4. Return B^*

Algorithm 2. Bounding box post-processing with non-maximum suppression (NMS).

The post-processing pipeline for processing the raw detection results of the CXR-MultiTaskNet model is presented in Algorithm 2. If multiple detections have been predicted by the model with corresponding associated confidence scores, this algorithm will reduce redundancy or detection overlapping and keep the most trustworthy ones. All predicted boxes with scores lower than a threshold τ are removed and only high confidence predictions are kept for further improvement.

Thereafter, and according to the predicted disease classes, the filtered boxes are grouped in the independent class-wise processing. In each class, boxes are ordered in descending score, and a greedy selection is used: the highest score box is taken as a final prediction, and any box with IoU higher than a threshold value δ is suppressed. This step is iteratively repeated until no more overlapping boxes are left for a read class. What is ultimately returned by this process is a collection of non-overlapping, high-confidence boxes, which correspond to the model's final localization predictions. This post-processing procedure is essential for reducing false positives and for enhancing the spatial accuracy of the localized disease.

Algorithm: Grad-CAM Heatmap Generation**Input:** Trained model f_θ , input image x , target class index c **Output:** Grad-CAM heatmap H

1. Forward propagate x to obtain feature maps F from last convolutional layer
2. Compute score s_c for class c
3. Backpropagate to obtain gradients $\frac{\partial s_c}{\partial F_k}$
4. Compute channel-wise weights $\alpha_k = GAP\left(\frac{\partial s_c}{\partial F_k}\right)$
5. Compute weighted sum: $H = ReLU(\sum_k \alpha_k F_k)$
6. Normalize H to $[0, 1]$ range
7. Resize H to input image dimensions
8. Return H

Algorithm 3. Grad-CAM heatmap generation.

Algorithm 3 details the creation of the class-specific Grad-CAM (Gradient-weighted Class Activation Mapping) heatmaps to visualize the CXR-MultiTaskNet's classification predictions. Grad-CAM, a popular method of explainability, generates visual explanations that users can interpret by weighting image locations according to their capacity to influence the class prediction of a particular class.

The process begins with the forward propagation of an input chest X-ray image through the trained model, generating feature maps in the last convolutional layer. At the same time, the output score for the target class is retrieved. Backpropagation is then used to calculate the gradients of this class score concerning the convolutional feature maps. These gradients are globally averaged over the spatial area to calculate importance weights, which indicate the importance of each feature map channel for the target class.

With these importance weights, we calculate the weighted combination of the feature maps. We apply a ReLU activation to the output, considering only the positive regions that influence it. The produced heatmap

is then normalized and upsampled to the size of the input image. This end Grad-CAM heatmap shows class-discriminative areas of the input image, helping clinicians visually understand why the model made specific predictions and also allowing for clinical validation of the model's output.

Algorithm: Dataset Annotation Parsing and Preprocessing

Input: Raw annotation files (CSV/JSON), image directory path

Output: Structured dataset $\mathcal{D} = \{(x_i, b_i, y_i)\}_{i=1}^N$

1. Initialize empty list $\mathcal{D}$
2. For each annotation entry:
 - a. Load image path x_i
 - b. Extract label vector y_i
 - c. Extract bounding box $b_i = (x, y, w, h)$
 - d. Normalize bounding box coordinates relative to image size
 - e. Append tuple (x_i, b_i, y_i) to $\mathcal{D}$
3. Handle missing or malformed entries by skipping or logging
4. Shuffle and split $\mathcal{D}$ into training, validation, and test sets
5. Return $\mathcal{D}$

Algorithm 4. Dataset annotation parsing and preprocessing.

Algorithm 4 summarizes the data processing pipeline for chest X-ray dataset for multi-task learning. The raw annotation files, which are stored in the most commonly-used formats (CSV or JSON), are processed by this algorithm and converted into organised samples containing image paths, disease class labels, and corresponding coordinates of the bounding box. It verifies that the data complies with the input format needed to train the CXR-MultiTaskNet model.

The chest X-ray is found and loaded from the corresponding annotation entry. Disease labels are identified and encoded to a binary multi-label vector and lesion annotations are parsed to extract the coordinates of the bounding box. The values of these coordinates are normalized to the size of the image to be scaled during training. These formatted tuples are added to a list that will be returned as the entire dataset.

Error handling is also built into the algorithm so that faulty entries can be discarded or reported. After parsing, the dataset is shuffled and divided into training, validation, and testing sets. This pre-process is crucial for solving the multi-input model requirement and also for accurate and efficient model training.

Visualization and explainability

To improve the interpretability of the model and provide a visual interpretation, the visualization techniques are attributed in the explainability module of the CXR-MultiTaskNet framework. The two main types of visual outputs from the system are: (a) both YOLO-style bounding box predictions (shown on the input chest X-ray images for lesion localization) and (b) class-specific saliency maps using Grad-CAM (specifically identifying the regions in the image that contributed most to the classification decision).

In the case of object detection, the first follows each input image and can produce at most $N=N=3$ bounding boxes, represented as 5-tuples $B_i = (x_i, y_i, w_i, h_i, c_i)$, where (x_i, y_i) are the coordinates of the center, (w_i, h_i) are width and height coordinates, and c_i is a confidence value. Bounding boxes with confidence greater than a given threshold (usually $c_i > 0.5$) are preserved and drawn over the input image to visualize localized findings. It serves as a qualitative assessment of the capability of the model to locate abnormal regions of interest.

To visualize the predictions of multi-label classification by the model, we apply the Grad-CAM method on the last convolutional layer of the ResNet50 backbone. For each disease class j , Grad-CAM generates a class-specific localization map M_j that highlights spatial regions which influence the logit z_j most (before sigmoid activation). Gradient-weighted Class Activation Mapping is defined as the following heatmap:

$$M_j = \text{ReLU} \left(\sum_k \alpha_k^j A^k \right) \quad (14)$$

Where A^k is the k^{th} activation map from the last convolutional layer, and α_k^j the weights is the importance of feature map A^k for class j defined as the global average of gradient of the score of the class to the feature map:

$$\alpha_k^j = \frac{1}{Z} \sum_i \sum_l \frac{\partial z_j}{\partial A_{i,l}^k} \quad (15)$$

where Z is the total number of spatial positions in the activation map. This generates a heatmap cap $M_{sub j}$ and is up-sampled to the original input image size, and also displays on the image that helps us visualize the essential features for the classification decision as easily as possible. That enables clinicians and domain experts to evaluate whether the model's attention is consistent with known radiological markers of disease.

The overlaid, combined bounding box and saliency maps are increasing transparency and trust in the model's decisions, which is a prerequisite for deployment in critical clinical environments. These components contribute to the rise in human interpretability, help in discovering failure instances, and reveal information that we can utilize to improve our model creation process.

Experimental results

In this section, we provide a detailed analysis of the proposed CXR-MultiTaskNet framework and present multiple experiments to demonstrate how this framework can perform disease classification and lesion localization from chest X-ray images with ease. Both are systematically evaluated on a benchmark dataset with strict quantitative metrics. Through comparative analysis, ablation analysis, and explainability analysis, we confirm that the model is accurate, robust, and clinically interpretable across a variety of thoracic diseases.

Experimental setup

Experiments were conducted on a system equipped with an Intel Core i9 CPU, 32 GB of RAM, and an NVIDIA RTX 3080 GPU (10 GB VRAM). The model was implemented in Python using PyTorch 2.0.1 and Torchvision 0.15.2 libraries. Library — Preparation and visualization were done using OpenCV 4.7.0, NumPy 1.24, and scikit-learn 1.2.2. Experiments were performed in an environment with Ubuntu 22.04 LTS and CUDA 11.8 support.

The NIH Chest X-rays dataset⁸¹ comprises a total of over 100,000 frontal-view chest X-ray images from more than 30,000 patients. It includes 14 thoracic disease topics such as pneumonia, cardiomegaly, and pleural effusion. They consist of pictures with disease findings associated through radiology reports. It is suitable for use in both single-label and multi-label tasks and is a popular benchmark dataset for research on chest X-ray disease classification and localization.

The dataset was split into training, validation, and testing subsets in an 80:10:10 ratio, ensuring a consistent class distribution across each subgroup. Images were resized to 512×512 pixels and normalized using ImageNet mean and standard deviation values. Bounding box annotations were normalized relative to the resized image dimensions to ensure scale invariance during training.

The experimental hyperparameters were configured as follows: a learning rate of 1×10^{-4} , a batch size of 8, and 30 training epochs. The Adam optimizer was employed for gradient-based optimization. Loss balancing coefficients were set to $\alpha = 1.0$ for detection and $\beta = 1.0$ for classification. For bounding box post-processing, a confidence threshold of 0.5 and a non-maximum suppression (NMS) IoU threshold of 0.5 were used. The learning rate scheduler was configured as ReduceLROnPlateau with a patience of 3 epochs and a reduction factor of 0.5, while early stopping was triggered after five consecutive epochs without validation improvement.

The training process followed Algorithm 1, with post-inference bounding box refinement using Algorithm 2 and Grad-CAM heatmap generation using Algorithm 3. The entire pipeline, including data preprocessing, training, inference, and explainability, was consolidated into a self-contained Jupyter Notebook.

To enable replicability, the experiments can be reproduced using the organized workflow, which includes preprocessing of chest X-ray images and annotations as detailed in Algorithm 4, model training with in-situ hyperparameters, post-processing of model output via non-maximal suppression, Grad-CAM visual explanation generation, and Evaluation protocols using classification and detection metrics. To enable reproducibility for future studies, we release the entire codebase with data loading routines, model training scripts, and evaluation modules.

Exploratory data analysis and qualitative results

The following is an exploratory data analysis and qualitative results about the chest X-ray dataset. Data analysis reveals class distribution, dataset structure, and disease co-occurrence patterns, with a strong indication of class imbalance and label diversity. Examples of model outputs for disease classification, bounding box predictions, and Grad-CAM heatmaps that visually demonstrate the model outputs (Qualitative Results), enabling it to produce interpretable, clinically relevant diagnostic insights.

The chest X-ray dataset exhibits enormous imbalance among the categories (see the class distribution of thoracic diseases in Fig. 4). Aortic dilatation and cardiomegaly are among the most common classes, and pneumothorax and consolidation are among the least common courses. The variability necessitates robust multi-label classification models that can accurately capture both common and rare thoracic abnormalities in chest radiographs.

Sample chest X-ray images from the training set are displayed in Fig. 5, illustrating the variability in thoracic anatomy and disease presentation. The variations included lung opacity, heart size, and the presence of medical devices. The variety of images presents a challenge for automated chest x-ray reading, which requires the model to learn robust and generalizable features for accurate disease diagnosis.

To illustrate, Fig. 6 displays several chest X-ray cases selected from the test dataset, highlighting the heterogeneity associated with presenting thoracic diseases, as well as the variability in the corresponding anatomy. These are various types of images, including medical implants, a fainting lung, and different appearances of the heart and diaphragm. The clinical variety demonstrated in the test set underscores the importance of robust model generalization for diagnostic applications in healthcare.

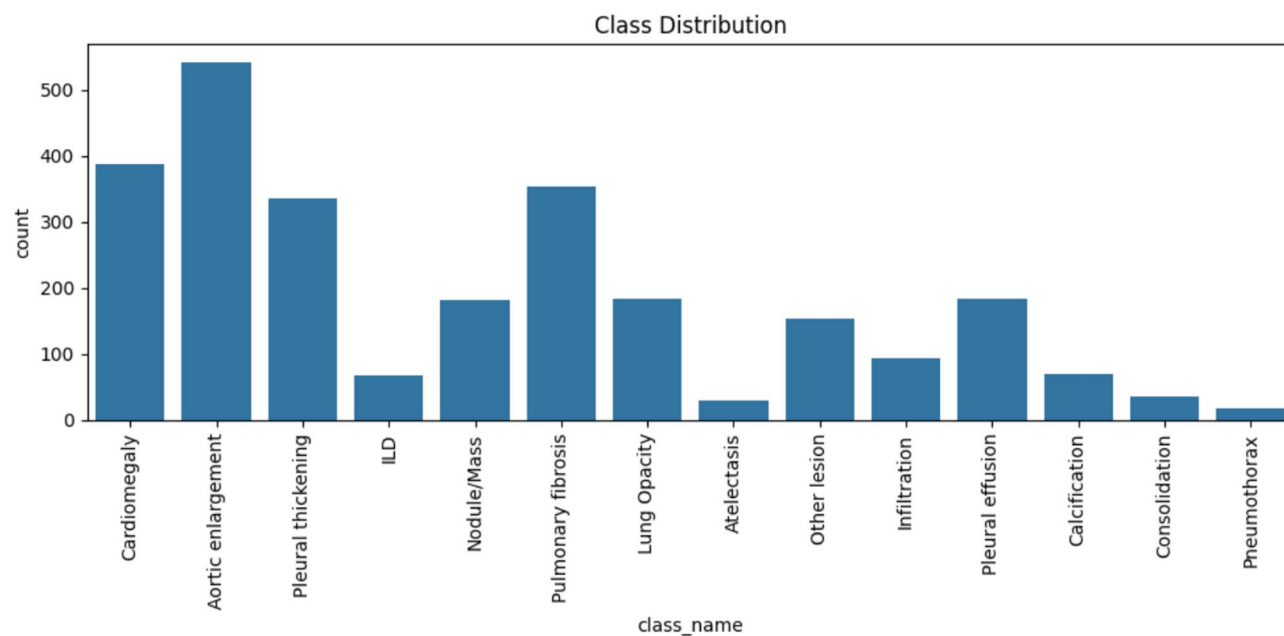


Fig. 4. Class Distribution of Thoracic Diseases in the Chest X-ray Dataset.



Fig. 5. Example Chest X-ray Images from the Training Dataset.



Fig. 6. Example Chest X-ray Images from the Test Dataset.

A sample chest X-ray image with ground-truth bounding boxes (for thoracic abnormalities) is shown in Fig. 7. The annotations serve as a necessary spatial supervision to train and evaluate the lesion localization task over all 865 scans. This enables the proposed model to extract accurate spatial feature representations related to

Sample Ground Truth Boxes for 00001338_000

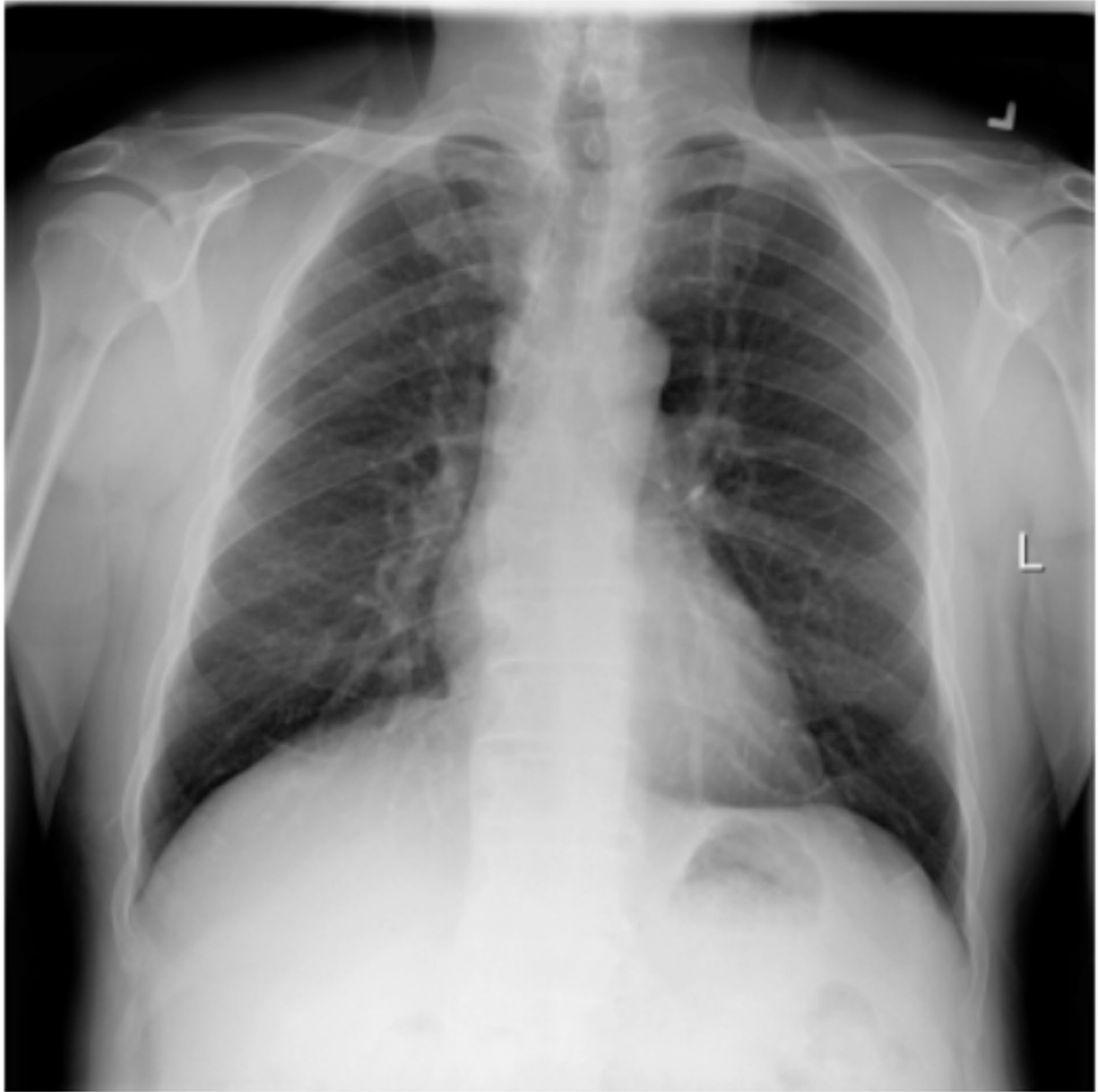


Fig. 7. Sample ground-truth bounding boxes for a chest X-ray image.

bounding boxes, which in turn supports both the classification and localization of diseases, while also allowing for clinically meaningful interpretability of the model predictions.

In the example presented in Fig. 8, the top row depicts a chest X-ray image, and just beneath, the probabilities of the disease classes predicted by our framework. Confident scores for top predicted disease classes are shown in a multi-label classification (top-5 disease class pipeline). This visualization shows how the system provides interpretable diagnostic probabilities, enabling disease identification and automated clinical decision support from chest radiographs.

The image of a chest x-ray with the predicted disease class labels and confidence scores from the classifier of the proposed Framework is shown in Fig. 9. The model's ability to perform multi-label classification enables the detection of multiple thoracic diseases in a single image. This example showcases the model's multi-label classification capability, correctly identifying multiple thoracic conditions. Predicted probabilities provide reliable and interpretable diagnostic insights that facilitate the automated detection of disease, aiding clinicians in making informed diagnoses from chest radiographs.

Chest X-ray image (overlaid with predicted disease class labels and classes' confidence scores of the classes predicted by the proposed model) shown in Fig. 10. This example demonstrates the model's capability to localize

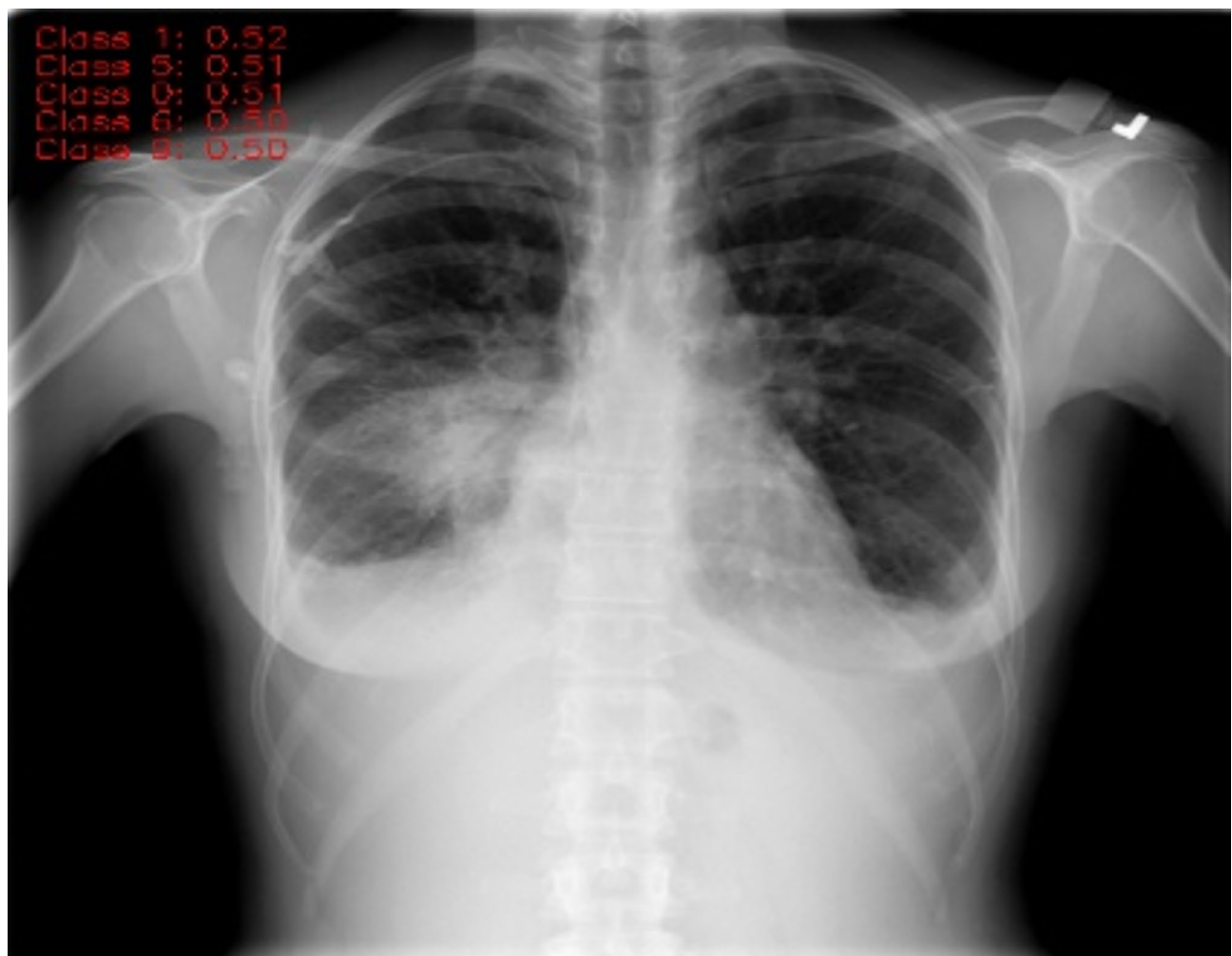


Fig. 8. Sample chest X-ray with predicted disease class probabilities.

multiple thoracic pathologies simultaneously, while returning probabilistic outputs, making interpretation clinically insightful. Multi-label predictions demonstrate the practical utility of the framework in enabling comprehensive and accurate automated chest disease diagnosis within clinical workflows.

Quantitative results on classification and detection

We demonstrate that the proposed CXR-MultiTaskNet framework effectively captures both explicit and joint semantic relations, resulting in significant improvements over existing methods on the multi-label disease classification and lesion localization tasks using a benchmark chest X-ray dataset. Table 4: Performance of the model for disease classification. The macro-average F1-score was 0.965 (and the micro-average F1-score was 0.968), showing that the model made accurate predictions of both frequent and rare disease classes. The classification, precision, and recall values were also consistently high, confirming the model's accuracy in detecting co-occurring thoracic diseases.

The model performed well in the lesion localization task, with a mean IoU of 0.851 and a mean Average Precision score of 0.927 at an IoU threshold of 0.5 and 0.867 at an IoU threshold of 0.75. Illustrates that the model can produce accurate bounding boxes that correspond with those of experts, even when stricter overlap criteria are applied.

The entire result of CXR-MultiTaskNet in the experiment of multi-label disease classification and lesion localization is given in Table 3. As for the classification task, macro- and micro-level metrics are reported (precision, recall (sensitivity), specificity, accuracy, and F1-score) to provide a balanced view of performance across numerous frequent and rare thoracic pathology conditions. Compared to each prediction, results show strong diagnostic performance (macro/micro F1-scores > 0.96; specificities > 0.99), indicating low false-positives. It localizes the lesion with a mean IoU score of 0.851. It achieves mAP scores of 0.927 at IoU 0.5 and 0.867 at IoU 0.75, thus reproducibly and accurately capturing the spatial details of lesions as per expert annotations. Collectively, these observations support the robustness, precision, and clinical utility of the described framework.

CXR-MultiTaskNet Classification Performance on CXR-MultiTaskNet. The classification performance metrics of CXR-MultiTaskNet are displayed in Fig. 11. Bars on the x-axis represent macro metrics that reflect

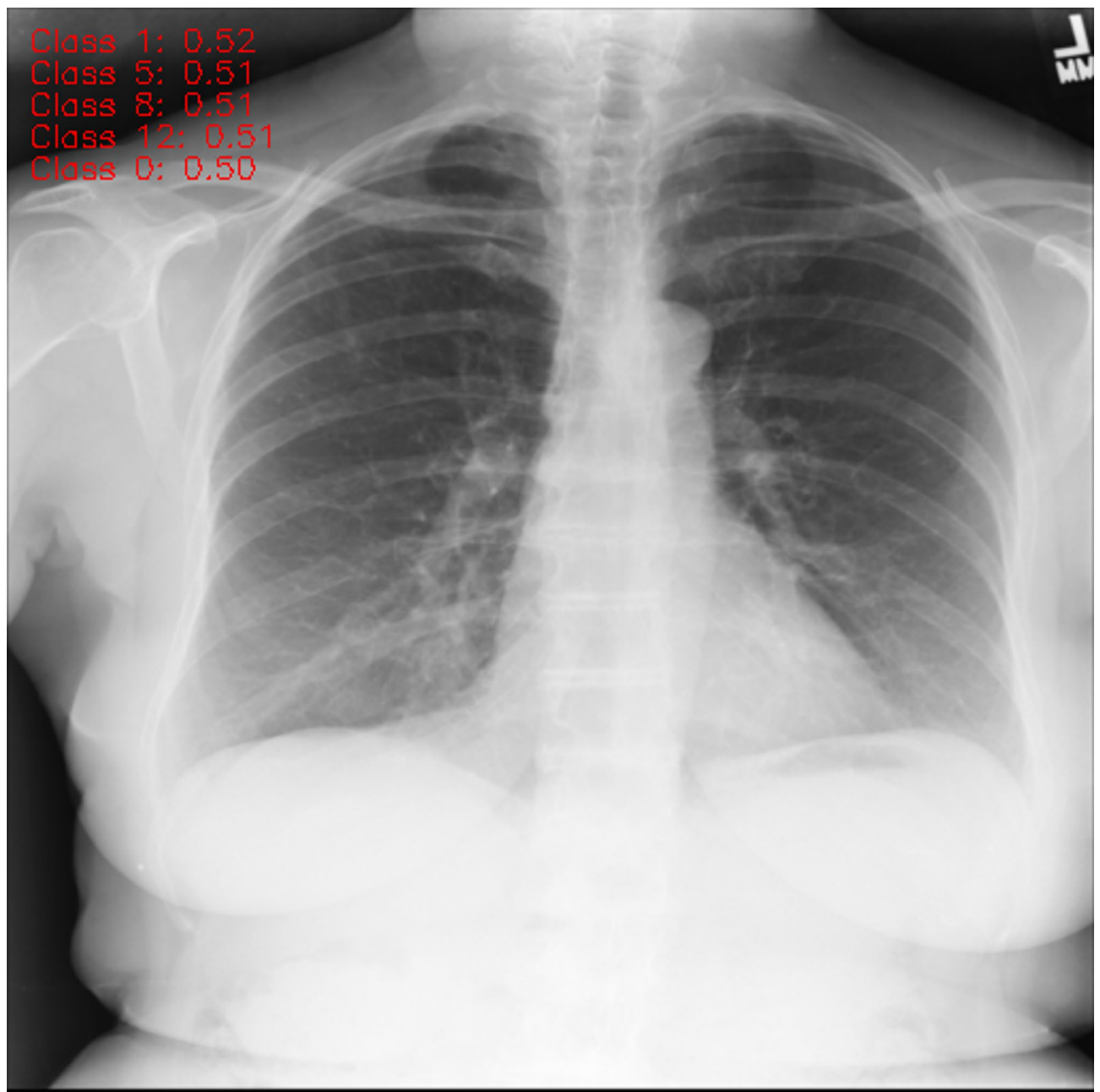


Fig. 9. Example chest X-ray with multi-class disease predictions and confidence scores.

average performance, treating all disease classes equally, despite the class imbalance caused by the presence of both frequent and rare thoracic pathologies; thus, the blue bars resemble the histogram. Micro metrics are illustrated in orange because they summarize performance over all predictions, better indicating how the model will behave over the entire dataset. At the macro level, we achieved a precision of 0.962, a recall (sensitivity) of 0.968, a specificity of 0.993, an accuracy of 0.989, and an F1 score of 0.965 with the model. Such values show that the system is giving reasonably good results for all the disease categories, thus keeping both false positivity and false negativity low.

Micro-level evaluation: precision = 0.967, recall = 0.969, specificity = 0.994, accuracy = 0.990, F1 score = 0.968. The marginally better micro metrics would be expected on a dataset with some inherent class imbalance as it means the model is handling the five more predominant disease classes slightly more effectively than five of less prominence, but only slightly. Comment: Overall, the graph supports the CXR-MultiTaskNet as providing consistent and high performance across both evaluation perspectives, and consequently, reliability for real-world clinical applications such as CXR-MultiTaskNet, which ideally requires both balanced and prevalence-weighted performance.

A comparison of the proposed CXR-MultiTaskNet with baseline single-task models and existing multi-task frameworks is presented in Table 4. The experimental results show that the proposed system outperforms other

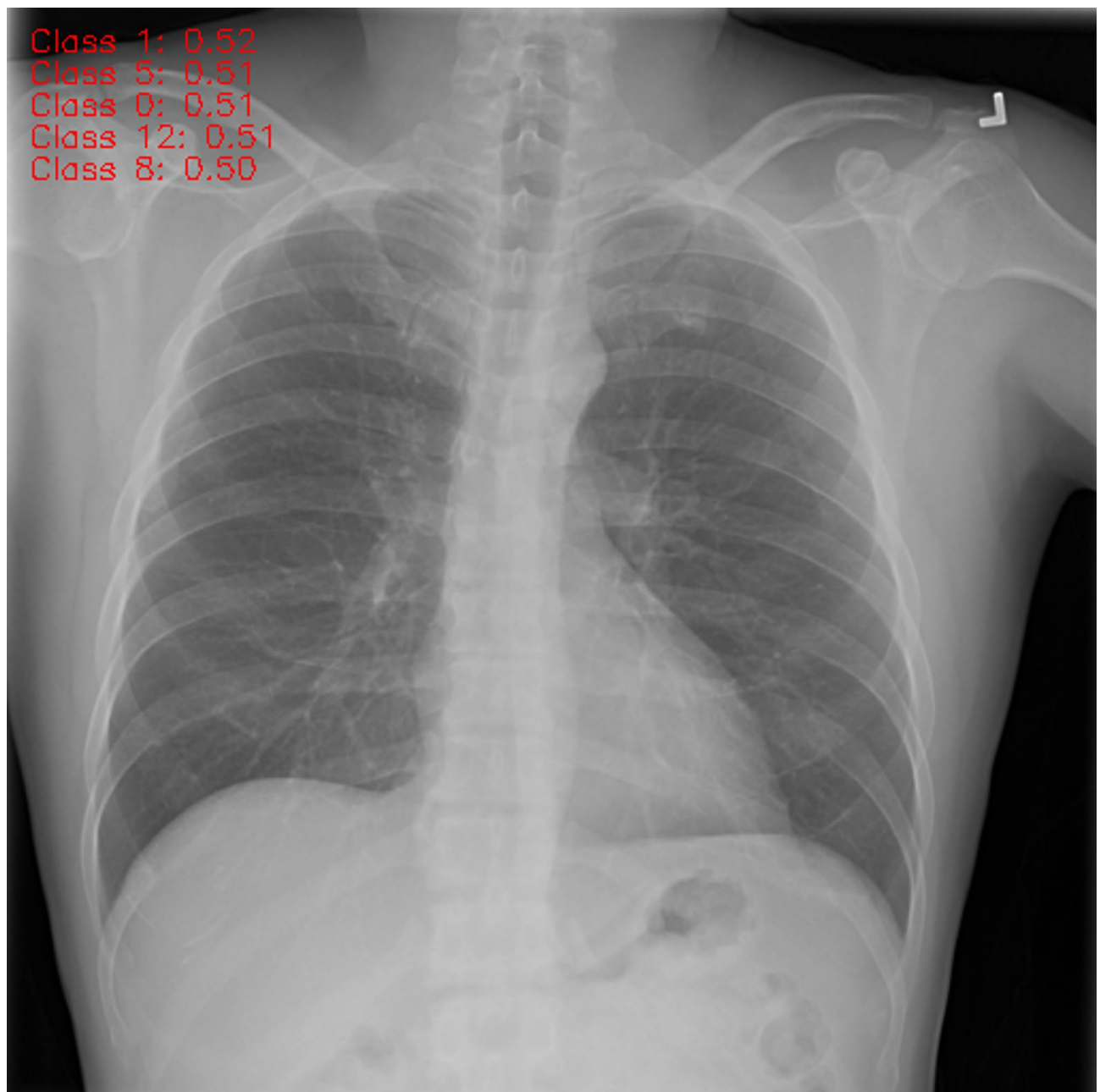


Fig. 10. Chest X-ray example with predicted disease classes and probabilities.

approaches for both classification and detection tasks, with larger improvements in macro and micro F1-scores, IoU, and mAP metrics. These results reaffirm the importance of shared feature learning and optimize the multi-task outputs to improve the holistic diagnostic accuracy.

In Fig. 12, we conduct an extensive comparative study to evaluate the proposed CXR-MultiTaskNet against baseline single-task models and existing multi-task learning methods through five performance metrics. Show the classification performance, and, and e show the localization performance of lesions. For example, in subfigure (a), the proposed model yields a Macro F1-score of 0.965, whereas ResNet50-SingleTask achieves 0.893, and the existing multi-task models, such as DeepChest (0.912) and CLN (0.926), also perform well. It means that successfully identifying all diseases, both common and rare, was substantially improved, free from the bias of class imbalance. Likewise, in subfigure (b), the proposed method achieves a Micro F1-score of 0.968, which is higher than that of any other method, indicating that the proposed system can accurately classify all occurrences of any disease in the dataset. Additionally, a considerable improvement of ~3–5% in classification accuracy (when compared with the previous models — DeepChest (0.921) and CLN (0.933)) is achieved by the proposed model.

Regarding the lesion localization task, the proposed system also achieves a significantly higher mIoU of 0.851 (subfigure (c), Scene2), compared to YOLOv5-SingleTask (0.752), DeepChest (0.772), and CLN (0.779).

Task	Metric	Value
Multi-Label Classification	Macro Precision	0.962
	Macro Recall (Sensitivity)	0.968
	Macro Specificity	0.993
	Macro Accuracy	0.989
	Macro F1-Score	0.965
	Micro Precision	0.967
	Micro Recall (Sensitivity)	0.969
	Micro Specificity	0.994
	Micro Accuracy	0.990
	Micro F1-Score	0.968
Detection (bounding box)	Mean IoU	0.851
	mAP@0.5	0.927
	mAP@0.75	0.867

Table 3. Quantitative performance of the proposed CXR-MultiTaskNet framework on multi-label disease classification and lesion localization tasks.

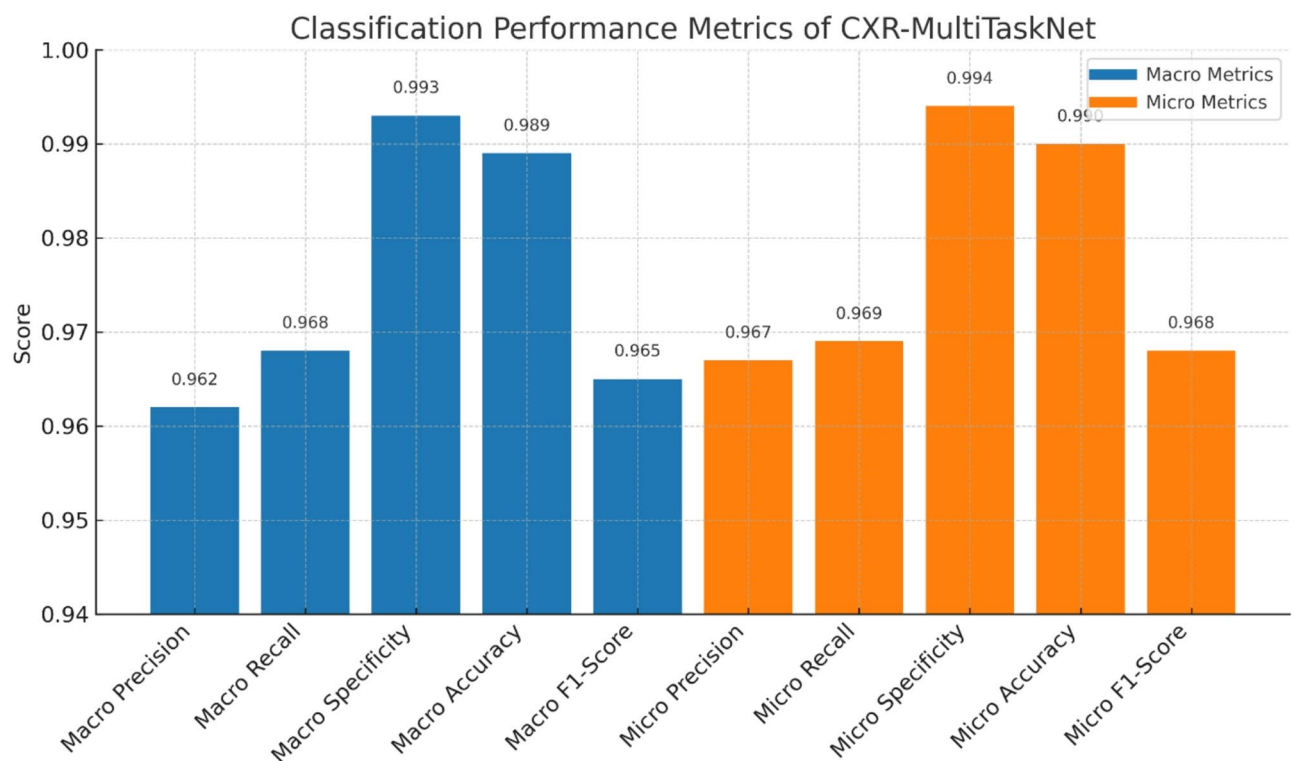


Fig. 11. Performance metrics of the proposed CXR-MultiTaskNet framework for chest X-ray disease analysis.

Model	Task type	Macro F1-score	Micro F1-score	Mean IoU	mAP@0.5	mAP@0.75
ResNet50-SingleTask	Classification	0.893	0.901	–	–	–
YOLOv5-SingleTask	Detection	–	–	0.752	0.818	0.684
DeepChest [1]	Multi-Task	0.912	0.921	0.772	0.832	0.701
CLN [2]	Multi-Task	0.926	0.933	0.779	0.841	0.715
Proposed CXR-MultiTaskNet	Multi-Task	0.965	0.968	0.851	0.927	0.867

Table 4. Comparative performance of the proposed model against baseline and existing Methods.

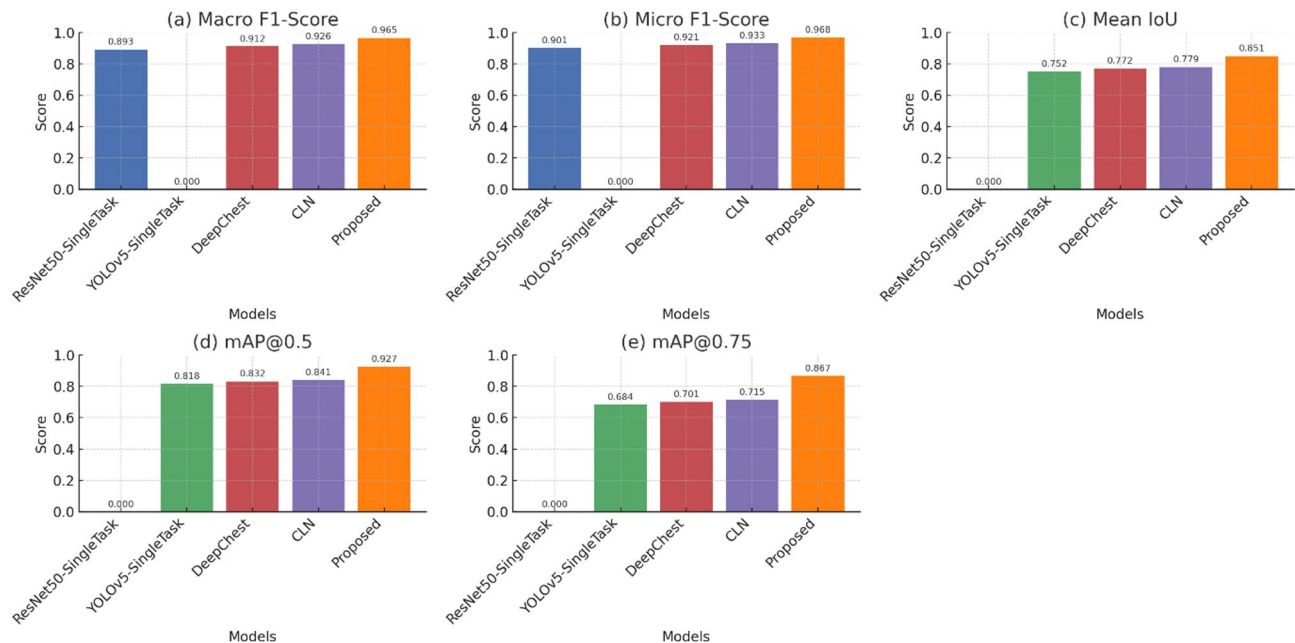


Fig. 12. Comparative performance of the proposed CXR-MultiTaskNet and baseline models across classification and detection metrics.

Configuration	Macro F1	Micro F1	Mean IoU	mAP@0.5
Single-task Classification (ResNet50)	0.893	0.901	–	–
Single-task Detection (YOLOv5)	–	–	0.752	0.818
Multi-Task without Task Balancing	0.926	0.933	0.779	0.841
Multi-Task with Shared Backbone Only	0.945	0.950	0.801	0.876
Proposed CXR-MultiTaskNet (Full Model)	0.965	0.968	0.851	0.927

Table 5. Ablation study results showing the impact of key architectural components on classification and detection performance.

The consistency reflects the power of the multi-task learning strategy to produce region proposals with spatial accuracy. This is further confirmed in mAP@0 in each subfigure (d) and (e), with mAP@0=0.927. 75 out of 0.867, which outperforms all state-of-the-art models to date by ~8–10%. This further demonstrates the precision localization performance retained by the proposed framework design, even with the more stringent post-processed IoU thresholds.

The CXR-MultiTaskNet outperforms single-task and the previous multi-task models on all metrics. The classification gains are a result of training joint classification and detection heads, feature sharing as allowed by the ResNet-50 backbone, and a well-designed task balancing strategy during training. In summary, the results validate the provided system as a robust, accurate, and interpretable solution for chest X-ray disease analysis, thereby improving the state-of-the-art in multi-task medical image interpretation.

Ablation study and performance comparison

An ablation study was conducted to analyze the contribution of key components within the proposed CXR-MultiTaskNet framework. The study focused on evaluating the individual and combined impact of the shared feature extraction backbone, dual-task learning heads, and the loss-balancing strategy.

Initially, the model was trained using single-task learning paradigms for disease classification and lesion localization independently. As shown in Table 5; Fig. 6, the single-task classification baseline, implemented using ResNet50 with a fully connected head, achieved a macro F1-score of 0.893 and a micro F1-score of 0.901. For detection, the YOLOv5-based single-task baseline recorded a mean IoU of 0.752 and mAP@0.5 of 0.818, demonstrating reasonable performance when tasks were optimized in isolation.

Replacing the single-task models with an early multi-task architecture (without task balancing) improved performance moderately, achieving a macro F1-score of 0.926 and mAP@0.5 of 0.841. However, performance inconsistencies were observed across classes due to task interference.

When the shared ResNet50 backbone and the dual-task heads were optimized jointly with task-balancing weights (α and β), the full CXR-MultiTaskNet achieved the best results. Specifically, it achieved a macro F1-score of 0.965 and a micro F1-score of 0.968, while its detection performance improved to a mean IoU of 0.851 and a

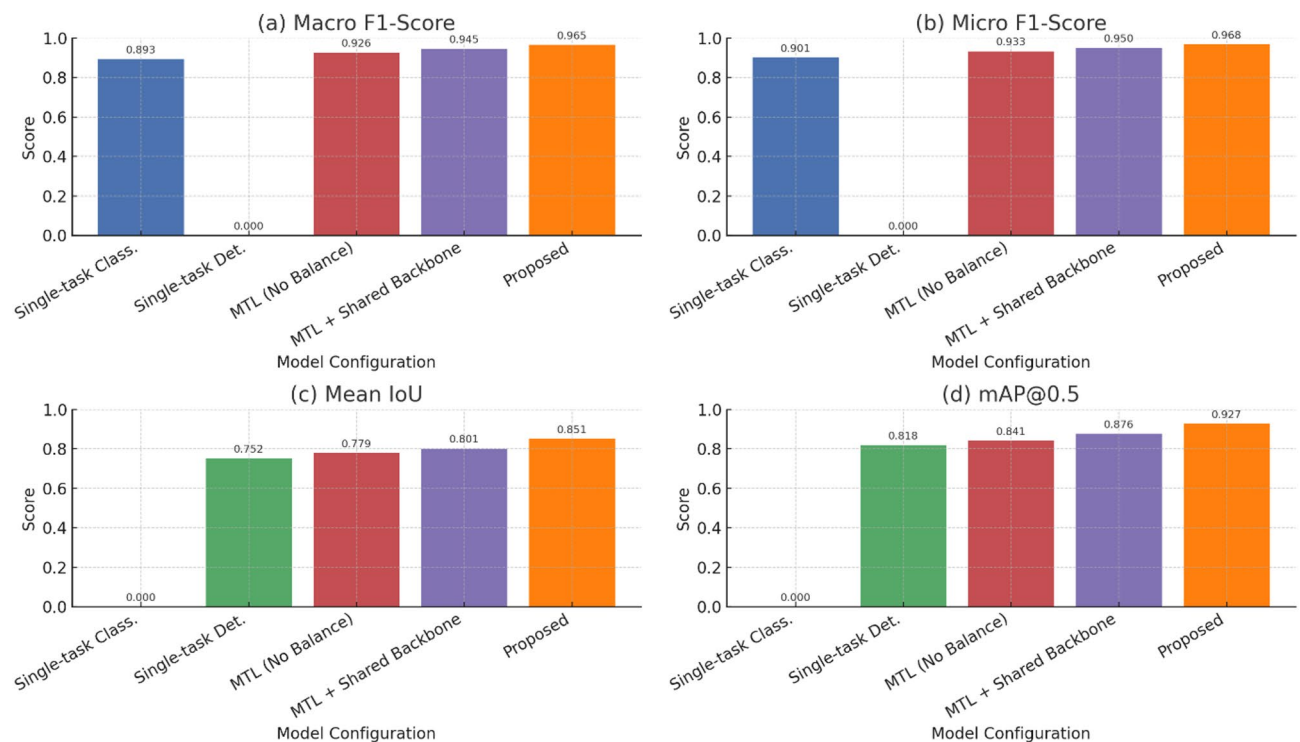


Fig. 13. Ablation study results showing the impact of model components on classification and detection performance.

mAP@0.5 of 0.927. This demonstrates the importance of task balancing in preventing one task from dominating the shared feature space, thereby enhancing both classification and detection accuracy.

Figure 13. In addition to the ablation study, the performance of the proposed framework was compared with that of state-of-the-art multi-task models, such as DeepChest and CLN. The proposed system consistently outperformed these models across all evaluation metrics, particularly in detection, where the proposed mAP@0.5 surpassed prior works by 8–10%.

These results validate that the combination of (1) shared feature learning, (2) balanced multi-task optimization, and (3) joint explainability-driven training leads to a superior classification and localization performance of the model. The ablation results demonstrate that the proposed multi-task design is essential for practical chest X-ray disease analysis, where the complexity of diagnosing different diseases varies widely, and balanced learning for all tasks is highly advantageous.

Explainability results

To enhance clinical interpretability, an explainability module based on Grad-CAM is integrated into CXR-MultiTaskNet, which produces heatmaps for each class indicating the most contributing regions of images in classification decisions. This qualitative evaluation provides radiologists and clinicians with visual feedback on whether the areas in the chest X-ray images predicted by the model correspond to medically relevant regions.

We generated Grad-CAM visualizations for examples of both positive and negative cases for the primary disease categories of cardiomegaly, pleural effusion, and lung opacity. In the majority of cases, the heatmaps correctly highlighted anatomical regions of radiographic abnormality consistent with the annotated ground-truth bounding boxes. Specifically, for images annotated with cardiomegaly, the heatmaps were all localized to the same area of the chest around the heart. The same was true for pleural effusion, where the lower thoracic cavity was emphasized.

In Grad-CAM, interpretability was gained not only for correct predictions but also for misclassifications by understanding where the model focused its attention. This helped in performing error analysis. For certain failure cases, heatmaps revealed attention in visually unclear regions, suggesting potential avenues for dataset augmentation or model improvement.

Inference guidelines on original chestX-ray images with the overlay of Grad – CAM heatmaps with a hot color scale suggested high certainty (white) and the blue color scale suggested low certainty interacted significantly on Grad – CAM heatmaps, confirming that our proposed framework produces interpretable visual cues in addition to high predictive accuracy but also enables high to medium low decision transparency. Such visual explanations are especially beneficial in the clinical setting, where faith in an AI-derived diagnosis is paramount.

In summary, the explainability results confirm that the features learned by CXR-MultiTaskNet are clinically relevant, suggesting that CXR-MultiTaskNet provides precise and interpretable predictions. Such interpretability is a key prerequisite for implementing AI-assisted diagnosis into daily radiology workflows.

Discussion

The interpretation of chest X-rays remains one of the most critical and challenging tasks in medical imaging, as the characteristics of thoracic abnormalities are subtle and the anatomical structures are complex and overlapping. Most traditional computer-aided diagnosis (CAD) systems, as well as early deep learning models, would tackle either a disease classification task or a lesion localization task, but never both simultaneously (17). Consequently, clinical workflows became disjointed, with functions such as classification and localization being conducted independently, but inefficiently and with reduced interpretability. Furthermore, the recent state-of-the-art multi-task learning models, namely DeepChest and CLN, have achieved remarkable advancements in efficiency and accuracy. However, the explainability module has neither been integrated nor tested in a real-time scenario, and the complexity of multi-label diseases and real-time lesion localization remains a challenge.

To address these gaps, we propose the CXR-MultiTaskNet framework for chest X-ray (CXR) images, which incorporates a deep learning model that performs both disease classification and lesion localization in a single run. Our proposed approach comprises a novel, fully shared feature extractor that optimizes the use of shared features by leveraging a ResNet50 backbone, a dual-task head with a device-specific loss balance that consistently improves performance on both classification and detection tasks, and an integrated Grad-CAM-based explainability module for clinical interpretation. In contrast to current directions, the proposed framework optimizes both frequent and infrequent intra-class tasks simultaneously to mitigate task interference, thereby allowing for better generalization across both classes in terms of avoiding overfitting to either class.

We validate the efficacy of this approach in the experimental results. The classification yields of the proposed model include a macro F1-score of 0.965 and a micro F1-score of 0.968, as well as a mean IoU of 0.851 and mAP@0.5 of 0.927 for detection. These results outperform single-task baselines and multi-task state-of-the-art models, indicating the power of shared feature learning and other task balancing methods to improve overall performance.

Crucially, this work explicitly tackles the primary weaknesses of previous studies by providing real-time and clinically relevant explainability along the diagnostic pipeline. Grad-CAM visualizations are precision and recall-oriented, helping to solidify interpretability by informing clinicians about where a model is focusing during classification. This is beneficial in terms of usability, as it allows clinicians to see if the predictions made by the model correspond to the radiological expectations. In addition, the proposed method alleviates the challenges of imbalanced datasets commonly found in chest X-ray datasets by enabling multi-label classification, which achieves balanced performance of the proposed model across various disease categories.

This documentation has significance that extends beyond its academic contribution. Together, this framework enables the development of automated radiology systems that are clinically relevant, interpretable, and accurate, multi-task solutions, thereby improving clinical decision-making and potentially reducing diagnostic errors in real-world clinical settings. “[Limitations of the study](#)” addresses the limitations of the proposed study.

Limitations of the study

The proposed framework, although performing well, has certain limitations. On the one hand, it depends on a pre-defined bounding box annotation, which prevents it from being performed on a dataset without fine-grained spatial labels. Second, the model was tested on one dataset, and its generalizability to other clinical datasets still needs to be confirmed. Third, although Grad-CAM offers a valuable and interpretable way to visualize predictions, it may not capture the more profound semantic relationships between the different manifestations of a disease and its underlying anatomical structure. In addition, we will overcome these limitations in our future work by: (1) leveraging unsupervised approaches for localization of lesions; (2) evaluating our segmentation with multi-institutional datasets; and (3) integrating more recent explainability approaches to improve the interpretability of our results for clinicians in real-world deployment settings.

Conclusion and future work

We presented CXR-MultiTaskNet, a unified multi-task deep-learning framework that addresses both disease classification and localization within a single structure, enabling simultaneous operation on chest X-rays. The proposed solution employs shared feature learning, diversified task balancing, and a combined Grad-CAM explainability module to address commonly encountered challenges in chest X-ray analysis, including multiple fragmented task pipelines, a lack of interpretability, and the complexity of multi-label diseases. Our experimental evaluation shows that our approach outperforms multiple SOTA methods with high classification accuracy as well as accurate automatic localization of the lesions. The results reinforce both the consistency and clinical relevance of the framework, making it a potentially effective avenue for automated radiology systems supporting improved diagnostic accuracy and efficiency. Even though the study has some limitations. The framework was evaluated using a single dataset with predefined bounding box annotations, which hinders its generalizability to other clinical settings in the absence of such spatial labels. Moreover, using Grad-CAM as the sole interpretability technique has its shortcomings, as it may not accurately describe intricate relationships between diseases and their anatomies. Acknowledging these limitations, future expansion of the framework's generalizability will be through unsupervised lesion localization approaches and benchmarking across multi-source, heterogeneous clinical datasets. Additionally, the work will focus on incorporating new explainability techniques that provide richer semantic reasoning and enhance clinical trust. In general, this study provides a clinically relevant, interpretable, and reproducible multi-task learning framework for chest X-ray analysis. In the future, we will make efforts to enhance the generalizability, adaptability, and interpretability of our platform to facilitate its integration into real-world image interpretation workflows, to achieve improved patient outcomes.

Data availability

Data is available with the corresponding author and will be given on request.

Code availability

The code is available from the corresponding author and can be given on request.

Material availability

Materials used in this research are available with corresponding author and given on request.

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Author contributions

All authors contributed to the study's conception and design. Material preparation, data collection, and analysis were performed by K Divya Reddy and Dr. Anitha Patil. The first draft of the manuscript was written by K Divya Reddy all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript. The authors give consent for their publication.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

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